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Supplemental Information

In Vivo RNA Interference Screens Identify Regulators of Antiviral CD4$^+$ and CD8$^+$ T Cell Differentiation

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Figure S1

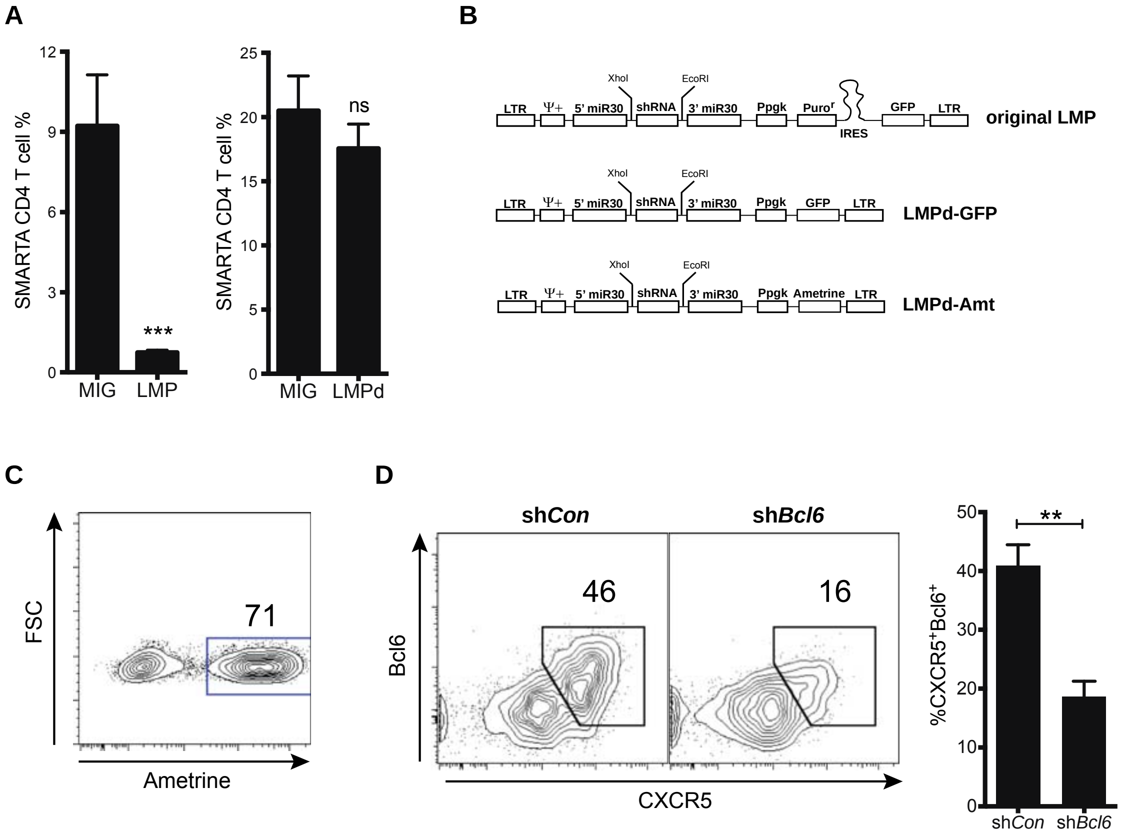
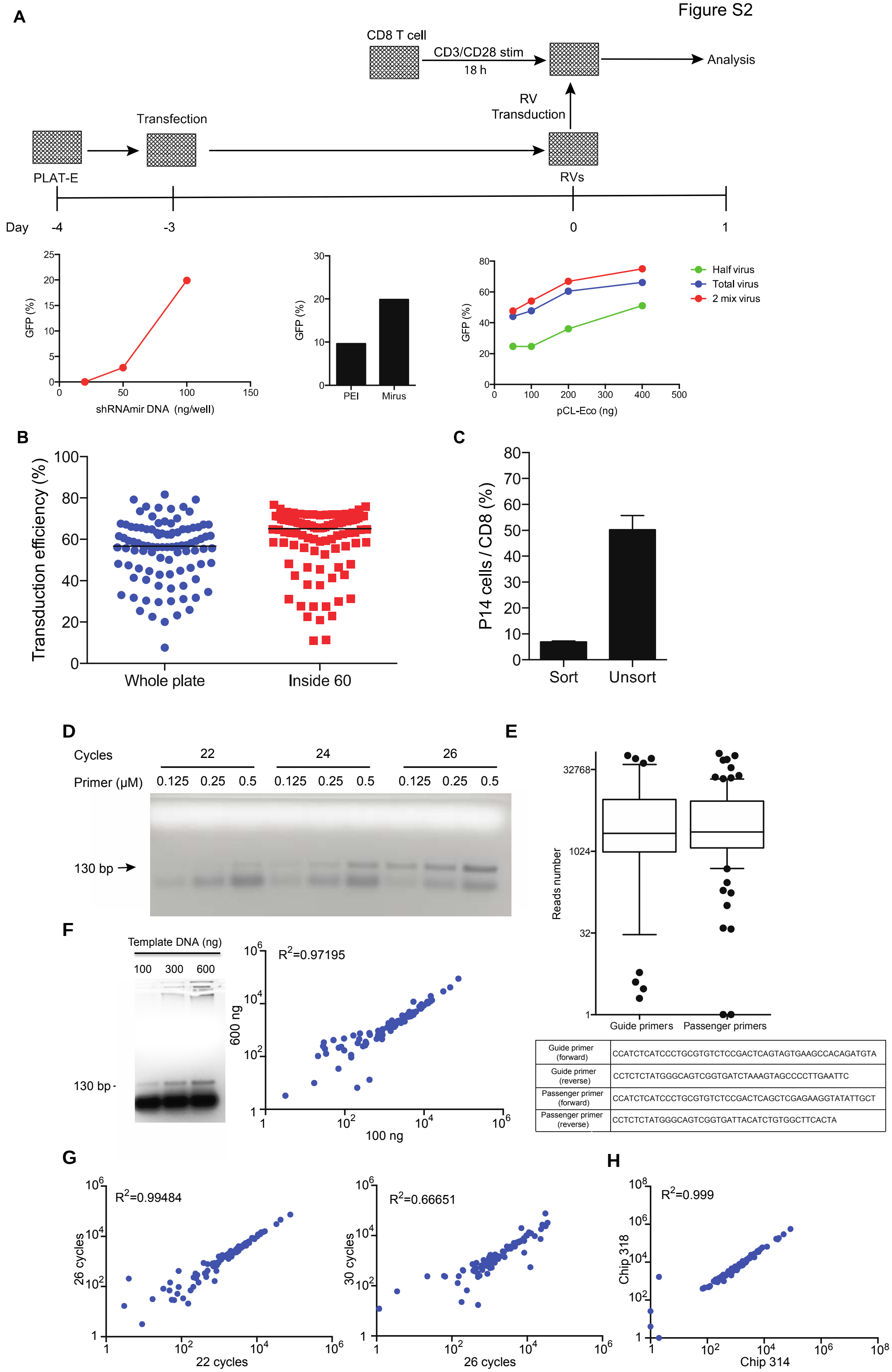


Figure S1: Modification of an shRNAmir-RV prevents rejection of transduced T cells after adoptive transfer and mediates efficient knockdown *in vivo* (related to Figure 1)**.** (**A** and **B**) SMARTA CD4 T cells were transduced with MIG-RV, LMP-RV or LMPd-RV, adoptively transferred into B6 mice, and analyzed 8 days after LCMV infection. (**A**) Quantitation shows SMARTA CD4 T cells as a percentage of total CD4 T cells. (**B**) Schematic representations of the LMP and LMPd vectors. (**C** and **D**) SMARTA CD4 T cells were transduced with LMPd-RV encoding *Con* or *Bcl6* shRNAmir, transferred into B6 mice and analyzed 6 days after LCMV infection. (**C**) Representative FACS plot of Ametrine (Ai et al., 2008) expression in transduced CD4 T cells. (**D**) Representative flow cytometry plots of CXCR5 and Bcl6 expression by SMARTA CD4 T cells transduced with sh*Con* or sh*Bcl6* shRNAmir-RV (left panel). $CXCR5^{+}Bcl6^{+}$ cells among total SMARTA CD4 T cells were quantified (right panel). *** P<0.001. Error bars indicate standard deviations.

Figure S2



Guide primer (forward)	CCATCTCATCCCTGCGTGTCTCCGACTCAGTAGTGAAGCCACAGATGTA
Guide primer (reverse)	CCTCTCTATGGGCAGTCGGTGATCTAAAGTAGCCCCTTGAATTC
Passenger primer (forward)	CCATCTCATCCCTGCGTGTCTCCGACTCAGCTCGAGAAGGTATATTGCT
Passenger primer (reverse)	CCTCTCTATGGGCAGTCGGTGATTACATCTGTGGCTTCACTA

Figure S2. Optimizations of retroviral transduction of CD8 T cells and of next generation sequencing libraries (related to Figure 1). **(A**) P14 CD8 T cells were activated in 96-well plates pre-coated with anti-CD3 and anti-CD28 antibodies. One day later, cells were transduced with empty shRNAmir-RV. Transduction efficiency was determined by assessing the frequency of GFP^+ cells by flow cytometry one day after transduction. P14 CD8 T cells were transduced with RV prepared by transfection of PLAT-E with increasing amounts of shRNAmir DNA (left), with different transfection reagents (middle) or with increasing amounts of the packaging vector pCL-Eco (right). In addition, P14 CD8 T cells were transduced with different volumes of RV supernatant (right). A representative experiment is shown out of two. (**B**) P14 CD8 T cells were plated in either a whole plate or in the inside 60 wells of 96-well plates coated with anti-CD3 and anti-CD28 antibodies followed with transduction with shRNAmir RV. One day after the transduction, the frequency of Ame^+ cells was determined by flow cytometry. (**C**) P14 CD8 T cells were activated in vitro with anti-CD3 and anti-CD28 followed by transduction. Transduced cells were either sorted into GFP^+ cells or left unsorted prior to transfer in B6 mice. The total number of splenic P14 and endogenous CD8 T cells were analyzed 5 days after infection with 1.5 x 10^5 PFU LCMV clone 13. (**D**) Sequencing libraries (130 bp) were generated by PCR amplification of proviral sequences in genomic DNA from pooled transduced cells using different primer concentrations and numbers of PCR cycles. (**E**) Sequencing reads number distribution in shRNAmir libraries from “input” generated with primers specific for the guide strand or for the passenger strand. (**F**) shRNAmir libraries (130 bp) were generated by PCR amplification from the indicated amounts of template DNA and were sequenced independently. The scatter plot of sequencing reads for each shRNAmir in libraries prepared with 100 ng or 600 ng of template DNA is shown. (**G**) The effects of extensive PCR is shown in scatter plots of sequencing reads for each shRNAmir in libraries prepared with 22, 26 or 30 cycles of PCR amplification. (**H**) Scatter plot of sequencing reads for each shRNAmir in a library sequenced with the Ion 314 Chip or with the Ion 318 Chip. The correlation coefficient, R^2, is indicated.

Figure S3

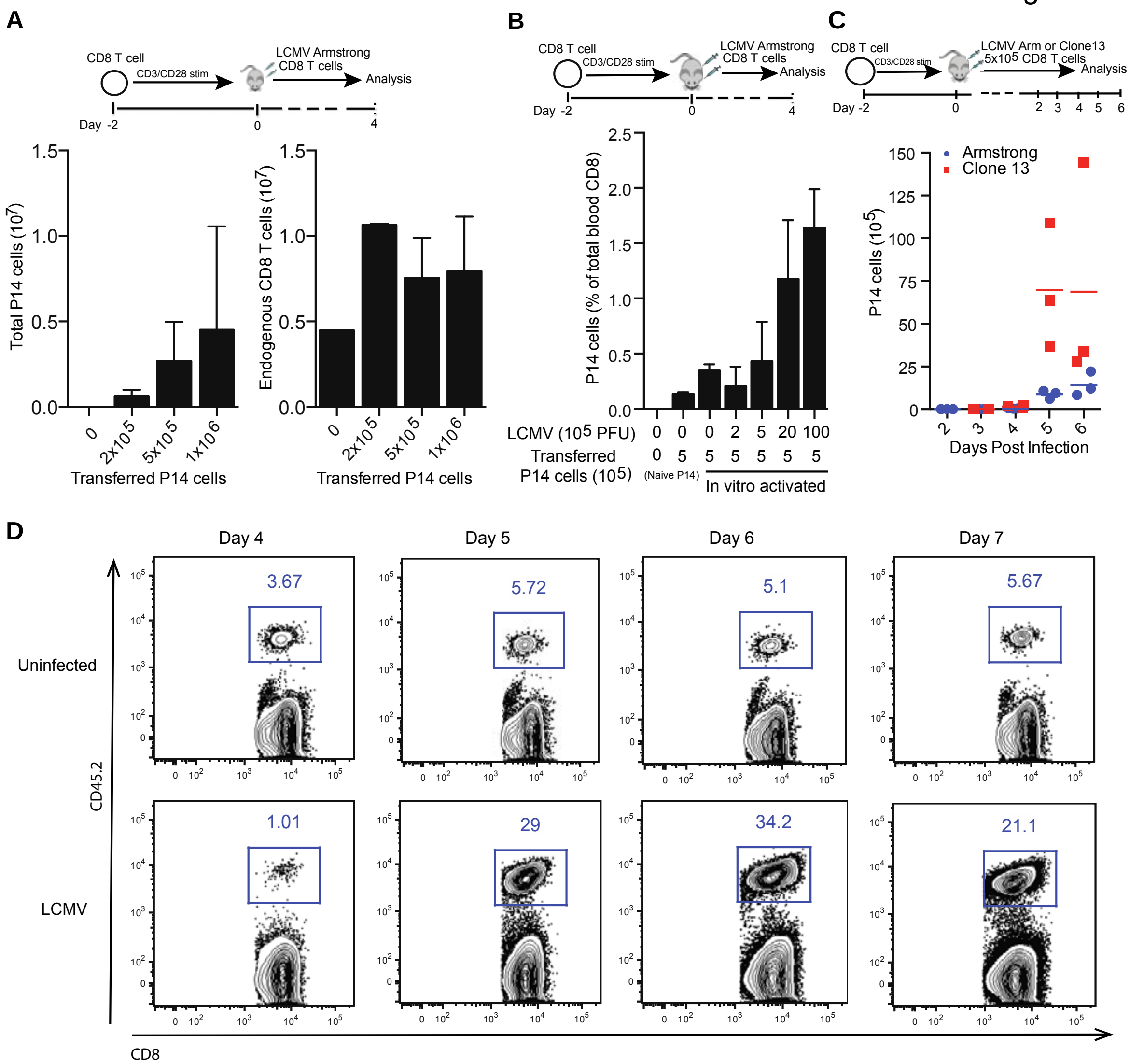


Figure S3. The number of transferred P14 cells as well as the strain and dose of LCMV was optimized for the CD8 T cell screen (related to Figure 1). (**A**) P14 CD8 T cells were activated in vitro with anti-CD3 and anti-CD28 for 2 days. Following activation, cells were transferred into B6.SJL mice, and immediately infected with $2x10^5$ PFU LCMV Armstrong 5. The total number of splenic P14 and endogenous CD8 T cells were analyzed 4 days after infection. Data shown as mean of two mice per group. (**B**) $P14^+$ $Thy1.1^+$ CD8 T cells were activated in vitro and transferred into B6 ($Thy1.2^+$) mice followed by infection with increasing doses of LCMV Armstrong 5. The number of P14 CD8 T cells in the blood was determined 4 days after infection. Data shown as mean ± SD of three mice per group. The percentage of P14 cells in the blood increased substantially in a fashion dependent on the increased dose of LCMV-Armstrong indicating that a stronger LCMV infection resulted in a larger P14 cell response. (**C**) $P14^+$ $Thy1.1^+$ CD8 T cells were activated in vitro and transferred into B6 ($Thy1.2^+$) mice. Mice were infected with LCMV Armstrong 5 ($1x10^7$ PFU) or LCMV clone 13 (1.5 $x10^5$ PFU). Splenic P14 CD8 T cell number was determined at different time points after infection.

Data shown as mean from three mice per group. P14 cells from LCMV clone 13 infected mice accumulated most strongly, suggesting that this infection was more robust than infection with high-dose LCMV Armstrong, and was less efficiently curtailed by the transfer of activated P14 cells (**D**) The frequency of $CD45.2^+$ P14 CD8 T cells was determined in the spleens of B6.SJL mice at the indicated time points following infection with 1.5×10^5 PFU LCMV clone13 or no infection. Representative dot plot graphs are shown. Error bars indicate standard deviations.

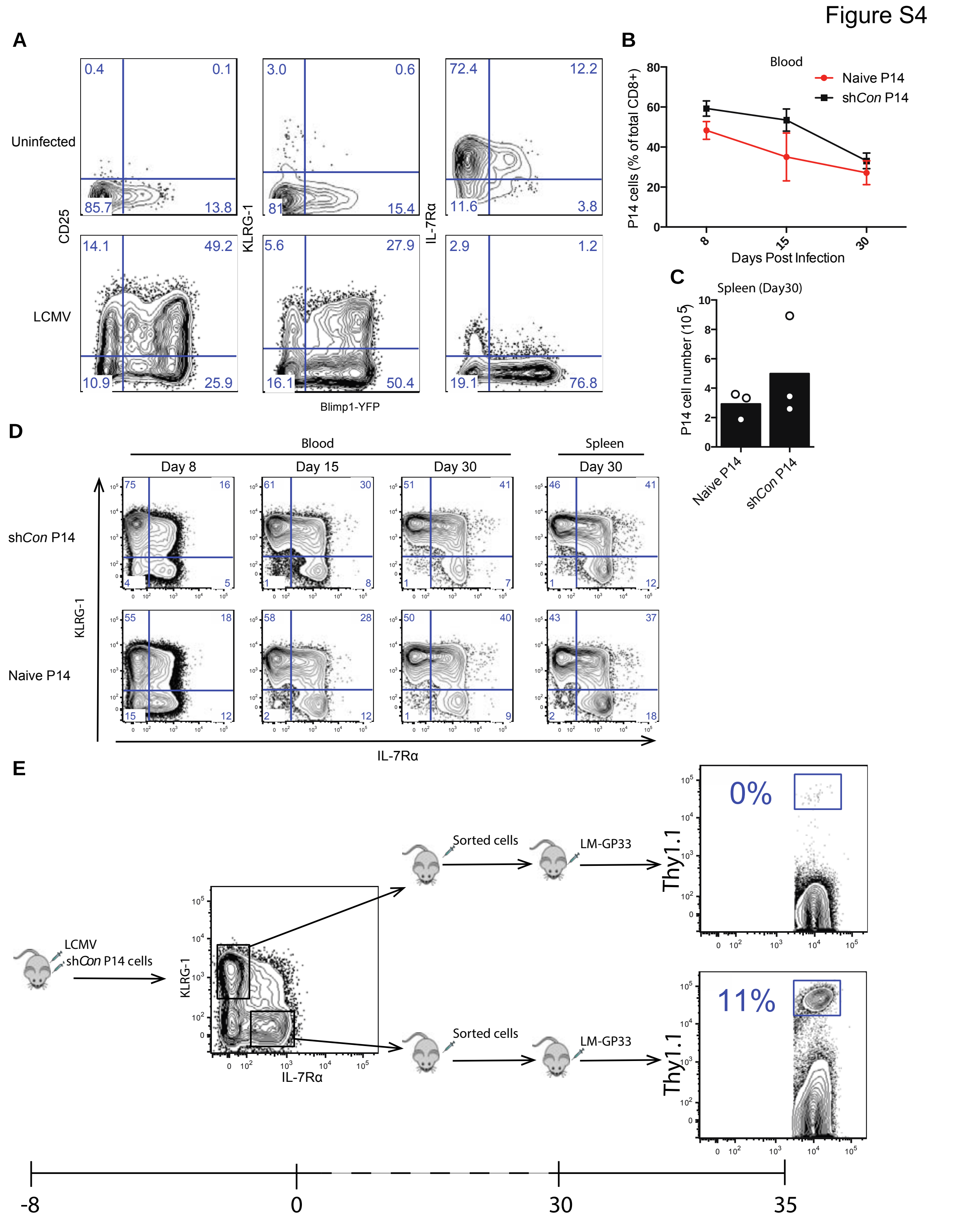


Figure S4. Activated and transduced P14 cells undergo effector CD8 T cell differentiation that is similar to naïve P14 cells following adoptive transfer and LCMV infection (related to Figure 1). **(A)** Flow cytometry shows staining of the indicated markers relative to Blimp1-YFP reporter expression of *in vitro*-activated and adoptively transferred P14 cells from representative infected and uninfected mice 5 days after transfer and infection. (**B**) Naïve P14 CD8 T cells (5,000) or P14 CD8 T cells transduced with sh*Con* (500,000) were adoptively transferred into B6 mice, and immediately infected with 1.5 x 10^5 PFU LCMV clone13. The number of P14 CD8 T cells in the blood was determined at different time points after infection. Similar percentages of P14 cells (~50-60%) were found in the blood and in the spleen on day 8 following infection in both groups. (**C**) The total number of splenic P14 CD8 T cells was analyzed 30 days after infection. Data shown as mean of three mice per group. (**D**) Representative contour plots of KLRG-1 and IL-7Rα expression in Naïve P14 CD8 T cells or P14 CD8 T cells transduced with sh*Con* at different time points after LCMV infection. P14 cell numbers from each group contracted over time, but were maintained comparably between both groups in these tissues until days 15 and 30. Both naïve, as well as activated and transduced, P14 cells resulted in selective accumulation of IL-7Rα^{hi} cells during transition into the memory phase; both groups showed a progressive decrease in the percentage of KLRG-1^{hi}IL-7Rα^{lo} cells, and a concomitant increase in both KLRG-1^{hi}IL-7Rα^{hi} and KLRG-1^{lo}IL-7Rα^{hi} cells during this time. (**E**) To confirm the functional potentials of these different subsets, KLRG-1^{hi}IL-7Rα^{lo} and KLRG-1^{lo}IL-7Rα^{hi} cells derived from P14 cells that had been activated and transduced with sh*Con*-RV prior to adoptive transfer and infection were sorted from mice on day 8 post infection, and transferred to naïve B6 recipients. P14 CD8 T cells were transduced with sh*Con*, transferred in 6-week-old B6 mice, and immediately infected with 1.5 x 10^5 PFU LCMV clone13. 8 days after infection, P14 CD8 T cells were isolated by positive selection using CD90.1 magnetic beads, and CD90.1$^+$ cells were sorted into effector cells (KLRG-1^{hi}IL-7Rα^{lo}) and memory precursor cells (IL-7Rα^{hi}KLRG-1^{lo}). Equal numbers ($7x10^5$) of effector cells or memory precursor cells were adoptively transferred into B6 mice. 30 days after transfer, mice were infected with $1x10^5$ CFU of LM-GP33. The percentage of P14 cells in total CD8 T cells were analyzed 5 days after infection. P14 cells derived from transfer of KLRG-1^{lo}IL-7Rα^{hi} expanded strongly by day 5 after LM-GP33 challenge, whereas P14 cells derived from transfer of KLRG-1^{hi}IL-7Rα^{lo} cells expanded weakly, confirming the appropriate functionality of the cells. Error bars indicate standard deviations.

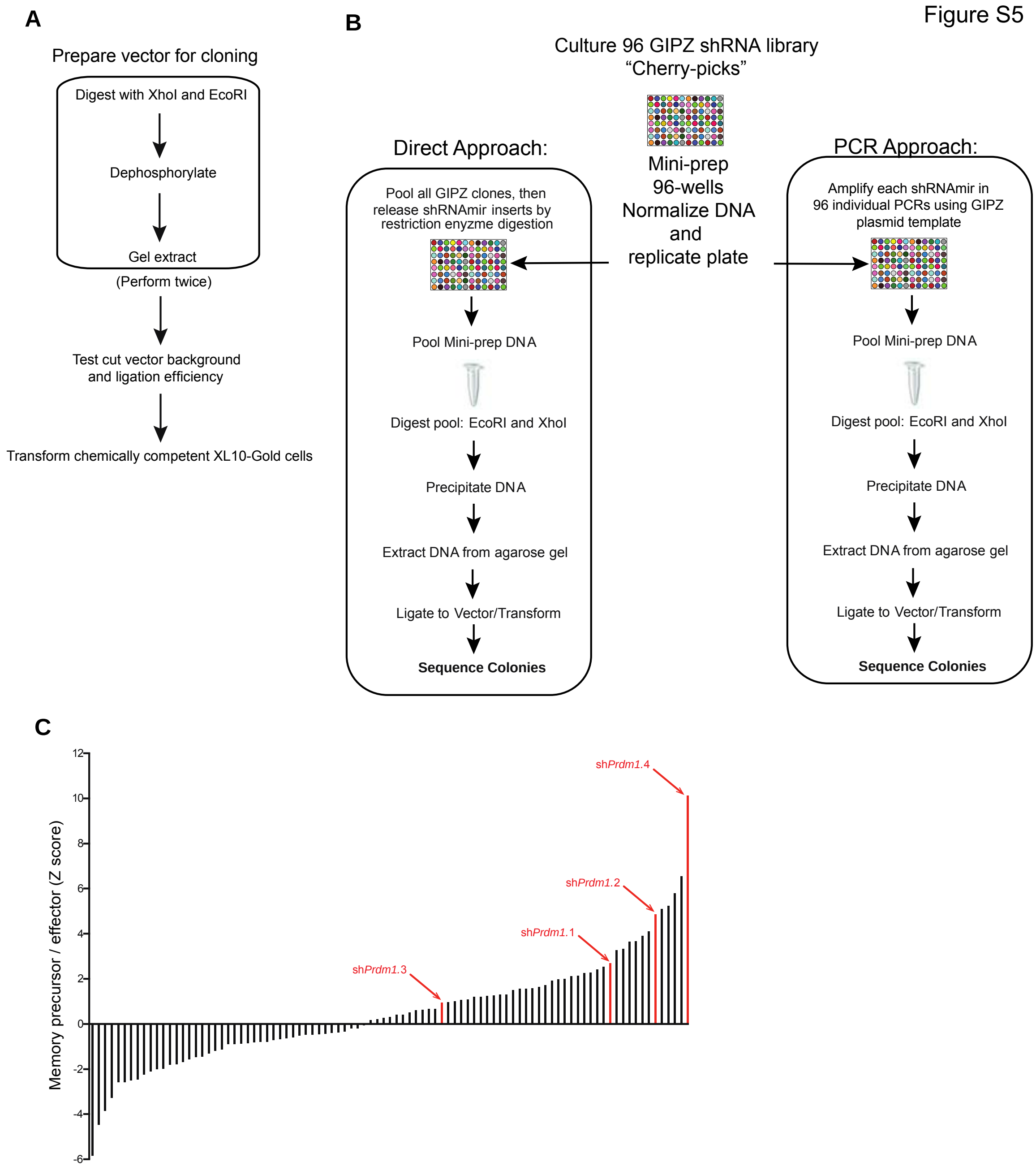


Figure S5. Cloning strategies to mobilize shRNAmir pools into LMPd (related to Figure 2). (**A**) LMPd-Ametrine vector was digested with EcoRI and XhoI twice. After purification, the cut vector background and ligation efficiency were tested. (**B**) 96 shRNAmirs DNA from GIPZ library were subcloned using two approaches: Direct and PCR Approaches. In the Direct approach, 96 shRNAmirs were pooled, digested with EcoRI and XhoI and purified. In the PCR approach, 96 shRNAmirs were individually amplified with PCR, then PCR products were pooled, digested with EcoRI and XhoI and purified. The purified shRNAmir fragments were ligated into EcoRI/XhoI-digested LMPd-Ametrine vector and transformed. 768 colonies were picked and sequenced. **(C)** Z-score values of all shRNAmirs in the first RNAi screen in CD8 T cells. shRNAmirs specific for *Prdm1* are highlighted in red.

Figure S6

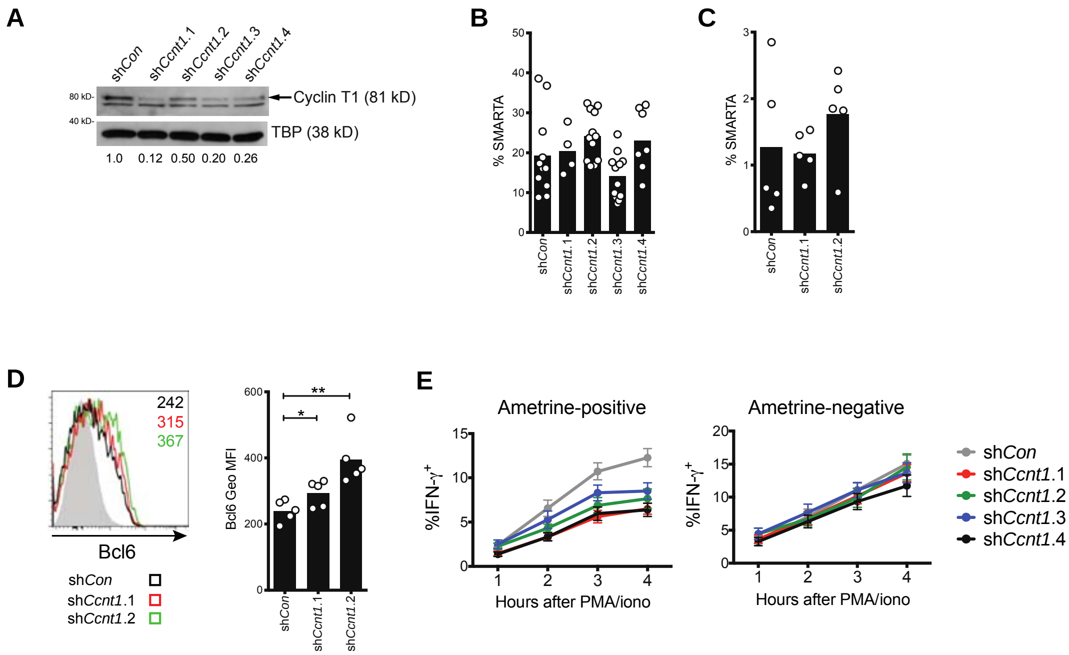


Figure S6. Ccnt1-specific shRNAs inhibit expression of Cyclin T1 and IFNγ and promote Tfh formation (related to Figure 4). (**A**) Expression of Cyclin T1 protein in WT CD4 T cells transduced with the indicated shRNAmir. Numbers below each lane indicate the ratio of Cyclin T1 to TBP relative to that of the control shRNAmir. (**B**) Quantitation of shRNAmir-transduced SMARTA CD4 T cells as a percentage of CD4 T cells 6 days after LCMV infection. (**C**) Quantitation of shRNAmir-transduced SMARTA CD4 T cells as a percentage of CD4 T cells 3 days after LCMV infection. (**D**) Bcl6 expression and geometric MFI quantitation in SMARTA transduced with the indicated shRNAmirs 4 days after LCMV infection. The numbers in the overlaid histograms indicate the geometric MFI of Bcl6 in SMARTA. (**E**) shRNAmir-transduced WT CD4 T cells were cultured under neutral conditions for 4 days before restimulation with PMA/ionomycin for the indicated times before staining for IFN-γ. Quantification of the IFN-γ^+ CD4 in the Ametrine-negative and –positive populations at the indicated times is shown. Each symbol represents T cells from an individual mouse. Data is representative of 2 independent experiments. *P<0.05, **P<0.01. Error bars indicate standard deviations.

Figure S7

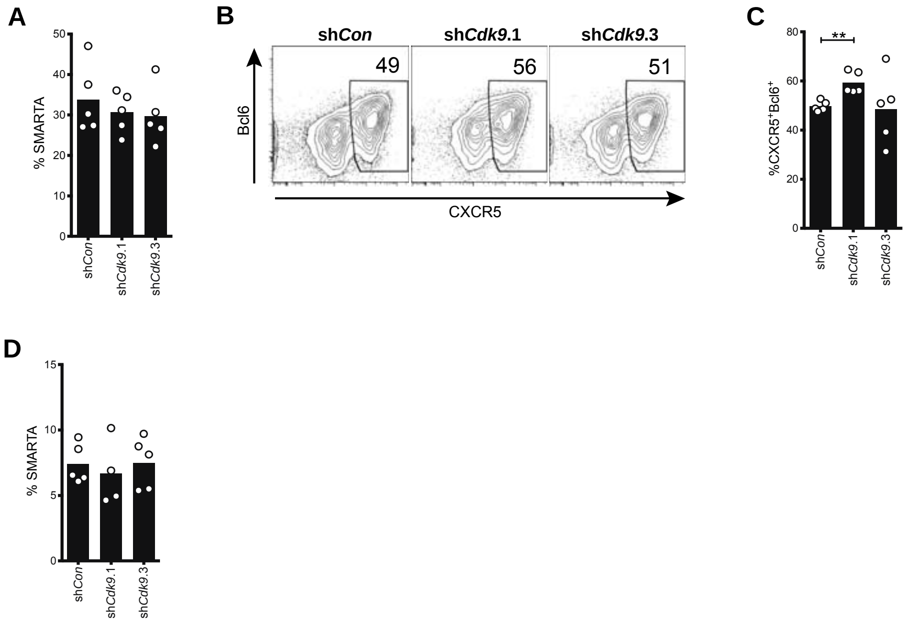


Figure S7. The P-TEFb subunit Cdk9 regulates Tfh formation (related to Figure 5). (**A**) Quantitation of shRNAmir-transduced SMARTA CD4 T cells as a percentage of CD4 T cells 6 days after LCMV infection. (**B**) Representative FACS plots of CXCR5 and Bcl6 expression in SMARTA CD4 T cells transduced with the indicated shRNAmir 6 days after LCMV infection. (**C**) Percentages of $CXCR5^{+}Bcl6^{+}$ Tfh in shRNA-transduced SMARTA cells. (**D**) Quantitation of shRNAmir-transduced SMARTA CD4 T cells as a percentage of CD4 T cells 4 days after LCMV infection. Data is representative of 2 independent experiments. **P<0.01, ***P<0.001.

Table S1. Table of individual shRNAmirs in the TF-30 library (related to Figure 2).

Mouse Gene Symbols	Full hairpin Sequence	CD4 Z-scores		CD8 Z-scores
Bcl6	TGCTGTTGACAGTGAGCGCGGCAAGTCCCTAATGAGTATATAGTGAAGCCACAGATGTATATACTCATTAGGGACTTGCCTTGCCTACTGCCTCGGA	-1.472404604	Bcl6	3.02060854
Bcl6.2	TGCTGTTGACAGTGAGCGCGCTGTCAAAGAGAAGGCTTTATAGTGAAGCCACAGATGTATAAAGCCTTCTCTTTGACAGCATGCCTACTGCCTCGGA	-3.636791478	Bcl6.2	0.626064286
Bcl6.3	TGCTGTTGACAGTGAGCGACGGCCTGTTCTACAGTATCTTTAGTGAAGCCACAGATGTAAAGATACTGTAGAACAGGCCGCTGCCTACTGCCTCGGA	-1.013977322	Bcl6.3	0.925845049
Bcl6.4	TGCTGTTGACAGTGAGCGCCGTTTCTCTGAGGAGGCCTATAGTGAAGCCACAGATGTATAGGCCTCCTCAGAGAAACGGTGCCTACTGCCTCGGA	ND	Bcl6.4	-1.750834964
Bcl6.5	TGCTGTTGACAGTGAGCGCGAGAGTCACTCACCACTCTAATAGTGAAGCCACAGATGTATTAGAGTGGTGAGTGACTCTCATGCCTACTGCCTCGGA	ND	Bcl6.5	2.84574501
Bcl6.6	TGCTGTTGACAGTGAGCGACAGCCTCTTATCCCATGTACATAGTGAAGCCACAGATGTATGTACATGGGATAAGAGGCTGTTGCCTACTGCCTCGGA	ND	Bcl6.6	-2.784446437
Ccnt1	TGCTGTTGACAGTGAGCGCCGACAGCCATTCTTCAGTTATTAGTGAAGCCACAGATGTAATAACTGAAGAATGGCTGTCGTTGCCTACTGCCTCGGA	4.613252583	Ccnt1	2.052768579
Ccnt1.2	TGCTGTTGACAGTGAGCGCCGATCAATACTGCTATAGTATTAGTGAAGCCACAGATGTAATACTATAGCAGTATTGATCGTTGCCTACTGCCTCGGA	2.089785808	Ccnt1.2	3.695203285
Ccnt1.3	TGCTGTTGACAGTGAGCGCCCTTGGAACTGTTAGATGAATTAGTGAAGCCACAGATGTAATTCATCTAACAGTTCCAAGGTTGCCTACTGCCTCGGA	0.352404369	Ccnt1.3	1.999916836
Ccnt1.4	TGCTGTTGACAGTGAGCGCGGAGATAAAGATGCGCATTAATAGTGAAGCCACAGATGTATTAATGCGCATCTTTATCTCCTTGCCTACTGCCTCGGA	3.825925218	Ccnt1.4	3.475527386
Cd14	TGCTGTTGACAGTGAGCGAGCCTCTTTGTTTAAGGAACATTAGTGAAGCCACAGATGTAATGTTCCTTAAACAAAGAGGCGTGCCTACTGCCTCGGA	0.612886034	Cd14	-0.027138234
Cd14.2	TGCTGTTGACAGTGAGCGCGGACTCAAGGTACTGAGTATTTAGTGAAGCCACAGATGTAAATACTCAGTACCTTGAGTCCATGCCTACTGCCTCGGA	0.093548446	Cd14.2	1.328975253
Cd19	TGCTGTTGACAGTGAGCGAATGGATAAGTCTGACGACCTATAGTGAAGCCACAGATGTATAGGTCGTCAGACTTATCCATGTGCCTACTGCCTCGGA	0.271001637	Cd19	-1.075677619
Cd19.2	TGCTGTTGACAGTGAGCGAAGGGACCTGGACTGTGACCTATAGTGAAGCCACAGATGTATAGGTCACAGTCCAGGTCCCTGTGCCTACTGCCTCGGA	-1.501913837	Cd19.2	ND
Cd19.3	TGCTGTTGACAGTGAGCGAAAGATGCAGACTCTTATGAAATAGTGAAGCCACAGATGTATTTCATAAGAGTCTGCATCTTCTGCCTACTGCCTCGGA	0.748833915	Cd19.3	-1.027207753
Cd19.4	TGCTGTTGACAGTGAGCGCCCCAGTGGTGACTTTAGTATATAGTGAAGCCACAGATGTATATACTAAAGTCACCACTGGGATGCCTACTGCCTCGGA	-1.04197917	Cd19.4	-1.434615984
Cd22	TGCTGTTGACAGTGAGCGAGCCCTCCCTGTCATCCTGAATTAGTGAAGCCACAGATGTAATTCAGGATGACAGGGAGGGCCTGCCTACTGCCTCGGA	1.421119184	CD22	ND
CD4	TGCTGTTGACAGTGAGCGAGCATGGGAGAAAGGATCGTTTTAGTGAAGCCACAGATGTAAAACGATCCTTTCTCCCATGCCTGCCTACTGCCTCGGA	ND	CD4	1.010696745
CD4.2	TGCTGTTGACAGTGAGCGACGGATGCAGAAGAGCCATAATTAGTGAAGCCACAGATGTAATTATGGCTCTTCTGCATCCGGTGCCTACTGCCTCGGA	ND	CD4.2	ND
CD4.3	TGCTGTTGACAGTGAGCGACCACTCAACTTTGCAGAGGAATAGTGAAGCCACAGATGTATTCCTCTGCAAAGTTGAGTGGGTGCCTACTGCCTCGGA	ND	CD4.3	0.94488072
CD4.4	TGCTGTTGACAGTGAGCGCCCTCTCATCATCAATAAACTTTAGTGAAGCCACAGATGTAAAGTTTATTGATGATGAGAGGATGCCTACTGCCTCGGA	ND	CD4.4	-0.52442329
CD4.5	TGCTGTTGACAGTGAGCGACACAGCTATCACGGCCTATAATAGTGAAGCCACAGATGTATTATAGGCCGTGATAGCTGTGCTGCCTACTGCCTCGGA	ND	CD4.5	ND
CD8	TGCTGTTGACAGTGAGCGGCATCAGATGTAACAAATCAATAGTGAAGCCACAGATGTATTGATTTGTTACATCTGATGCTTGCCTACTGCCTCGGA	-1.650641847	CD8	ND
CD8.2	TGCTGTTGACAGTGAGCGCCTCCGAATCTTTCCAAAGAAATAGTGAAGCCACAGATGTATTTCTTTGGAAAGATTCGGAGTTGCCTACTGCCTCGGA	0.139746385	CD8.3	ND
Cebpa	TGCTGTTGACAGTGAGCGAGGCCAGACCACCATGCACCTATAGTGAAGCCACAGATGTATAGGTGCATGGTGGTCTGGCCGTGCCTACTGCCTCGGA	2.206802784	Cebpa	8.733733627
Cebpa.2	TGCTGTTGACAGTGAGCGCCGGCCGCTGGTGATCAAACAATAGTGAAGCCACAGATGTATTGTTTGATCACCAGCGGCCGTTGCCTACTGCCTCGGA	1.158715616	Cebpa.2	0.942346369
Chd4.1	TGCTGTTGACAGTGAGCGACCCGATCGTGGTGGAAGACAATAGTGAAGCCACAGATGTATTGTCTTCCACCACGATCGGGGTGCCTACTGCCTCGGA	12.35263213	Chd4	-0.811294231
Chd4.2	TGCTGTTGACAGTGAGCGAAGGGAGAAGACAGCAGTGTTATAGTGAAGCCACAGATGTATAACACTGCTGTCTTCTCCCTCTGCCTACTGCCTCGGA	11.13189705	Chd4.2	3.389340759
Cul	TGCTGTTGACAGTGAGCGTGGAAGGAATGTTTAGGGATATTAGTGAAGCCACAGATGTATAACACTGCTGTCTTCTCCCTCTGCCTACTGCCTCGGA	-0.452556073	Cul3	9.182896459
Cul3.1	TGCTGTTGACAGTGAGCGAACAACTTTCTTCAAACACTAATAGTGAAGCCACAGATGTATAACACTGCTGTCTTCTCCCTCTGCCTACTGCCTCGGA	-1.478545867	Cul3.1	-0.049541074
Fos	TGCTGTTGACAGTGAGCGCGGAAAGTGAATTTGAATGAAATAGTGAAGCCACAGATGTATTTCATTCAAATTCACTTTCCATGCCTACTGCCTCGGA	0.216527228	Fos	1.713204427
Fosb	TGCTGTTGACAGTGAGCGCTGCCAGGAACCAGCTACTCAATAGTGAAGCCACAGATGTATTGAGTAGCTGGTTCCTGGCATTGCCTACTGCCTCGGA	-8.520194116	Fosb	-4.283736605
Fosb.2	TGCTGTTGACAGTGAGCGACTGCAAGATCCCCTACGAAGATAGTGAAGCCACAGATGTATCTTCGTAGGGGATCTTGCAGCTGCCTACTGCCTCGGA	0.254522092	Fosb.2	-3.753112494
Hes1.1	TGCTGTTGACAGTGAGCGCCCGATCACTAAGTAGCCCTAATAGTGAAGCCACAGATGTATTAGGGCTACTTAGTGATCGGTTGCCTACTGCCTCGGA	-1.566610567	Hes1	5.949368093
Hes1.2	TGCTGTTGACAGTGAGCGCGGATTGCGCCTTTGTATTATATAGTGAAGCCACAGATGTATATAATACAAAGGCGCAATCCATGCCTACTGCCTCGGA	-0.642904109	Hes1.2	1.156895696
Hes1.3	TGCTGTTGACAGTGAGCGACGAAGGGCAAGAATAAATGAATAGTGAAGCCACAGATGTATTCATTTATTCTTGCCCTTCGCTGCCTACTGCCTCGGA	0.356405022	Hes1.3	-0.580170111
Hes1.4	TGCTGTTGACAGTGAGCGCCACGACACCGGACAAACCAAATAGTGAAGCCACAGATGTATTTGGTTTGTCCGGTGTCGTGTTGCCTACTGCCTCGGA	-2.054491443	Hes1.4	2.573473318
Hes1.5	TGCTGTTGACAGTGAGCGAACAGAAAGTCATCAAAGCCTATAGTGAAGCCACAGATGTATAGGCTTTGATGACTTTCTGTGTGCCTACTGCCTCGGA	-1.732209459	Hes1.5	-0.34531361
Hes1.6	TGCTGTTGACAGTGAGCGCTGGATGCACTTAAGAAAGATATAGTGAAGCCACAGATGTATATCTTTCTTAAGTGCATCCAATGCCTACTGCCTCGGA	2.789152974	Hes1.6	-3.451456414
Hes1.7	TGCTGTTGACAGTGAGCGCCTGACGGACACTAAAAACGAATAGTGAAGCCACAGATGTATTCGTTTTTAGTGTCCGTCAGATGCCTACTGCCTCGGA	0.454912902	Hes1.7	-0.637726515
Hes6.1	TGCTGTTGACAGTGAGCGCCGATGCTCCAACCGAATTAAATAGTGAAGCCACAGATGTATTTAATTCGGTTGGAGCATCGATGCCTACTGCCTCGGA	0.292336008	Hes6	-1.585275942
Hes6.2	TGCTGTTGACAGTGAGCGAATCGATGCTCCAACCGAATTATAGTGAAGCCACAGATGTATAATTCGGTTGGAGCATCGATGTGCCTACTGCCTCGGA	-0.090954247	Hes6.2	-1.326093304
Hes6.3	TGCTGTTGACAGTGAGCGAAACAGATGACTTGAATGACAATAGTGAAGCCACAGATGTATTGTCATTCAAGTCATCTGTTGTGCCTACTGCCTCGGA	0.461060872	Hes6.3	-3.924947982
Id2.1	TGCTGTTGACAGTGAGCGCGCAAAGTACTCTGTGGCTAAATAGTGAAGCCACAGATGTATTTAGCCACAGAGTACTTTGCTTGCCTACTGCCTCGGA	0.549057486	Id2	-4.255480314
Id2.2	TGCTGTTGACAGTGAGCGCGCAGCACGTCATCGATTACATTAGTGAAGCCACAGATGTAATGTAATCGATGACGTGCTGCATGCCTACTGCCTCGGA	9.878826913	Id2.2	0.245813458
Id2.3	TGCTGTTGACAGTGAGCGCCTGAGCTTATGTCGAATGATATAGTGAAGCCACAGATGTATATCATTCGACATAAGCTCAGATGCCTACTGCCTCGGA	4.522581352	Id2.3	3.524126674
Id2.4	TGCTGTTGACAGTGAGCGCATGTCGAATGATAGCAAAGTATAGTGAAGCCACAGATGTATACTTTGCTATCATTCGACATATGCCTACTGCCTCGGA	0.835547328	Id2.4	0.117217827
Id2.5	TGCTGTTGACAGTGAGCGCCCGGTGAGGTCCGTTAGGAAATAGTGAAGCCACAGATGTATTTCCTAACGGACCTCACCGGATGCCTACTGCCTCGGA	-2.707032275	Id2.5	4.223759934
Id3.1	TGCTGTTGACAGTGAGCGCGCCCTGATTATGAACTCTATATAGTGAAGCCACAGATGTATATAGAGTTCATAATCAGGGCATGCCTACTGCCTCGGA	1.424922008	Id3	-2.417911199
Id3.2	TGCTGTTGACAGTGAGCGCGCCCTCTTCACTTACCCTGAATAGTGAAGCCACAGATGTATTCAGGGTAAGTGAAGAGGGCTTGCCTACTGCCTCGGA	-0.099771909	Id3.2	-1.377577609
Id3.3	TGCTGTTGACAGTGAGCGACCTGATTATGAACTCTATAATTAGTGAAGCCACAGATGTAATTATAGAGTTCATAATCAGGGTGCCTACTGCCTCGGA	-0.624767862	Id3.3	-3.067275625
Itch.1	TGCTGTTGACAGTGAGCGCGGGAACTGCTGGATTAGATATTAGTGAAGCCACAGATGTAATATCTAATCCAGCAGTTCCCATGCCTACTGCCTCGGA	-6.698554012	Itch	-3.362235094
Itch.2	TGCTGTTGACAGTGAGCGCCACAGTAGATGGACAGTCAAATAGTGAAGCCACAGATGTATTTGACTGTCCATCTACTGTGATGCCTACTGCCTCGGA	-3.772740403	Itch.2	1.475969582
Itch.3	TGCTGTTGACAGTGAGCGAAGGGAGCAATGCAGCAGTTTATAGTGAAGCCACAGATGTATAAACTGCTGCATTGCTCCCTGTGCCTACTGCCTCGGA	-7.184263289	Itch.3	-1.473784367
Itch.4	TGCTGTTGACAGTGAGCGGTCAAACAATATGAAACTTGAATAGTGAAGCCACAGATGTATAACACTGCTGTCTTCTCCCTCTGCCTACTGCCTCGGA	-3.684139828	Itch.4	-0.843332614
Klf12.1	TGCTGTTGACAGTGAGCGAGCCCAAGAACTGAGACTTATTTAGTGAAGCCACAGATGTAAATAAGTCTCAGTTCTTGGGCCTGCCTACTGCCTCGGA	0.440588967	Klf12	0.326647027
Klf12.3	TGCTGTTGACAGTGAGCGATCCGAGGAAATCGAATGAATATAGTGAAGCCACAGATGTATATTCATTCGATTTCCTCGGACTGCCTACTGCCTCGGA	-0.399229982	Klf12.3	-2.312904924
Klf12.4	TGCTGTTGACAGTGAGCGCACCGAGCTTGTGGAATCTGAATAGTGAAGCCACAGATGTATTCAGATTCCACAAGCTCGGTTTGCCTACTGCCTCGGA	2.577743884	Klf12.4	4.912189103
Klf12.5	TGCTGTTGACAGTGAGCGATCCCATGAACTTACAGTCTAATAGTGAAGCCACAGATGTATTAGACTGTAAGTTCATGGGACTGCCTACTGCCTCGGA	-1.109917896	Klf12.5	6.122233268
Klf2.1	TGCTGTTGACAGTGAGCGACCTTGCAAACAGACTGCTATTTAGTGAAGCCACAGATGTAAATAGCAGTCTGTTTGCAAGGGTGCCTACTGCCTCGGA	-2.405407802	Klf2	4.800951039
Klf2.2	TGCTGTTGACAGTGAGCGCCACCAAGAGCTCGCACCTAAATAGTGAAGCCACAGATGTATTTAGGTGCGAGCTCTTGGTGTTGCCTACTGCCTCGGA	8.862553743	Klf2.2	8.886670645
Klf2.3	TGCTGTTGACAGTGAGCGACGGCACGGATGAGGACCTAAATAGTGAAGCCACAGATGTATTTAGGTCCTCATCCGTGCCGCTGCCTACTGCCTCGGA	-6.781915592	Klf2.3	3.711852126
Klf2.4	TGCTGTTGACAGTGAGCGAAAGACAGGAGTCTGTGAAGAATAGTGAAGCCACAGATGTATTCTTCACAGACTCCTGTCTTCTGCCTACTGCCTCGGA	-3.335849379	Klf2.4	4.741839305
Klf2.5	TGCTGTTGACAGTGAGCGCATGACCGATTGTATTTCTATATAGTGAAGCCACAGATGTATATAGAAATACAATCGGTCATATGCCTACTGCCTCGGA	-7.799932852	Klf2.5	0.81214814
Klf2.6	TGCTGTTGACAGTGAGCGCCAGGTGAGAAGCCTTATCATATAGTGAAGCCACAGATGTATATGATAAGGCTTCTCACCTGTTGCCTACTGCCTCGGA	ND	Klf2.6	2.371815981
Klf2.7	TGCTGTTGACAGTGAGCGACCACACATACTTGCAGCTACATAGTGAAGCCACAGATGTATGTAGCTGCAAGTATGTGTGGCTGCCTACTGCCTCGGA	ND	Klf2.7	-1.613589553
Klf6.1	TGCTGTTGACAGTGAGCGCCGTGTCAGCCTCTGAGATAAATAGTGAAGCCACAGATGTATTTATCTCAGAGGCTGACACGTTGCCTACTGCCTCGGA	-0.605697224	Klf6	-2.242556924
Klf6.2	TGCTGTTGACAGTGAGCGCCAGAACTGAAGATTTCTTCTATAGTGAAGCCACAGATGTATAGAAGAAATCTTCAGTTCTGATGCCTACTGCCTCGGA	0.976666008	Klf6.2	-1.863456707
Mafb.1	TGCTGTTGACAGTGAGCGACCGGCCAAACCATCTTGAGAATAGTGAAGCCACAGATGTATTCTCAAGATGGTTTGGCCGGGTGCCTACTGCCTCGGA	-0.390637861	Mafb	-2.311194227
Mafg.1	TGCTGTTGACAGTGAGCGAGGCTCTCAGCTTAATCCAGAATAGTGAAGCCACAGATGTATTCTGGATTAAGCTGAGAGCCCTGCCTACTGCCTCGGA	-1.294074097	Mafg	-2.187626984
Mafg.2	TGCTGTTGACAGTGAGCGCGGTTATATAGCCCTTTATTTATAGTGAAGCCACAGATGTATAAATAAAGGGCTATATAACCATGCCTACTGCCTCGGA	-0.828253692	Mafg.2	-4.644658132
Ms4a1.1	TGCTGTTGACAGTGAGCGCGGATTTGACACCATATGAATATAGTGAAGCCACAGATGTATATTCATATGGTGTCAAATCCATGCCTACTGCCTCGGA	0.848541302	Ms4a1	-0.071971441
Ms4a1.2	TGCTGTTGACAGTGAGCGCGCAAGCCAAGATATTCAATAATAGTGAAGCCACAGATGTATTATTGAATATCTTGGCTTGCTTGCCTACTGCCTCGGA	-0.155474207	Ms4a1.2	0.876481604
Mta3	TGCTGTTGACAGTGAGCGCCAGACGGATAGAGGAGCTCAATAGTGAAGCCACAGATGTAATAATTAATAGCAACAAACGGCTGCCTACTGCCTCGGA	-3.86979074	Mta3	-3.753112494
Mta3.2	TGCTGTTGACAGTGAGCGAGGCCGTTTGTTGCTATTAATTTAGTGAAGCCACAGATGTAAATTAATAGCAACAAACGGCCGTGCCTACTGCCTCGGA	2.819365816	Mta3.2	-1.327510406
Mta3.3	TGCTGTTGACAGTGAGCGACCGTTTGTTGCTATTAATTATTAGTGAAGCCACAGATGTAATAATTAATAGCAACAAACGGCTGCCTACTGCCTCGGA	-0.976975959	Mta3.3	-0.900965492
Mta3.4	TGCTGTTGACAGTGAGCGACCACTCCCTGTAGGCCAAGTATAGTGAAGCCACAGATGTATACTTGGCCTACAGGGAGTGGCTGCCTACTGCCTCGGA	-2.838719318	Mta3.4	2.329001888
Mta3.6	TGCTGTTGACAGTGAGCGACGCCATCAGCAAAGACCCAAATAGTGAAGCCACAGATGTATTTGGGTCTTTGCTGATGGCGGTGCCTACTGCCTCGGA	-1.064662318	Mta3.6	-2.211503977
Nfatc2	TGCTGTTGACAGTGAGCGACGAAATCAGACCTAGAAACAATAGTGAAGCCACAGATGTATTGTTTCTAGGTCTGATTTCGGTGCCTACTGCCTCGGA	-1.797398994	Nfatc2	-0.576390203
Nfatc3	TGCTGTTGACAGTGAGCGCCGAGTAAGCTCAATGGTTAAATAGTGAAGCCACAGATGTATTTAACCATTGAGCTTACTCGTTGCCTACTGCCTCGGA	ND	Nfatc3	ND
Nfatc3.2	TGCTGTTGACAGTGAGCGCGGTGATATTTCTGAATCTAAATAGTGAAGCCACAGATGTATTTAGATTCAGAAATATCACCATGCCTACTGCCTCGGA	0.212378437	Nfatc3.2	-0.740341683
Nfatc3.3	TGCTGTTGACAGTGAGCGACACGACGAGCTCGACTTCAAATAGTGAAGCCACAGATGTATTTGAAGTCGAGCTCGTCGTGGTGCCTACTGCCTCGGA	-1.02669336	Nfatc3.3	-7.998229268
Nfatc3.5	TGCTGTTGACAGTGAGCGAGGCACTCCAGTGCTGAAGTATTAGTGAAGCCACAGATGTAATACTTCAGCACTGGAGTGCCGTGCCTACTGCCTCGGA	ND	Nfatc3.5	ND
Nfatc3.7	TGCTGTTGACAGTGAGCGAGCCCAGATCACAGGTCAGCCTTATAGTGAAGCCACAGATGTATAAGGCTGACCTGTGATCTGTGCCTACTGCCTCGGA	ND	Nfatc3.7	12.89520498
Nfatc3.8	TGCTGTTGACAGTGAGCGACCCAGGACATCTCTTAGCACATATAGTGAAGCCACAGATGTATATGTGCTAAGAGATGTCCTTGCCTACTGCCTCGGA	ND	Nfatc3.8	-9.268318025
Plagl1.1	TGCTGTTGACAGTGAGCGACCAATTATTCTTCAGGAGCATTAGTGAAGCCACAGATGTAATGCTCCTGAAGAATAATTGGGTGCCTACTGCCTCGGA	1.348938985	Plagl1	-0.324856408
Plagl1.2	TGCTGTTGACAGTGAGCGCGGAGCTGATGCAAGAGAATATTAGTGAAGCCACAGATGTAATATTCTCTTGCATCAGCTCCTTGCCTACTGCCTCGGA	-4.774232288	Plagl1.2	0.61873606
Pou2af1.1	TGCTGTTGACAGTGAGCGCGGATATGGTATGGAGGATAAATAGTGAAGCCACAGATGTATTTATCCTCCATACCATATCCATGCCTACTGCCTCGGA	0.049254973	Pou2af1	-4.5211207
Pou2af1.2	TGCTGTTGACAGTGAGCGAGGCCATCAGCTCCCTGACCATTAGTGAAGCCACAGATGTAATGGTCAGGGAGCTGATGGCCCTGCCTACTGCCTCGGA	2.101205446	Pou2af1.2	15.15493936
Prdm1.1	TGCTGTTGACAGTGAGCGCCGGAGCCATGAATCTCATTAATAGTGAAGCCACAGATGTATTAATGAGATTCATGGCTCCGTTGCCTACTGCCTCGGA	16.6736162	Prdm1	1.082696612
Prdm1.2	TGCTGTTGACAGTGAGCGAGCAGAGACTGGGATCATGAAATAGTGAAGCCACAGATGTATTTCATGATCCCAGTCTCTGCCTGCCTACTGCCTCGGA	-0.246585485	Prdm1.2	2.687468725
Prdm1.3	TGCTGTTGACAGTGAGCGAGGACTCCTACTCCTACTTGAATAGTGAAGCCACAGATGTATTCAAGTAGGAGTAGGAGTCCCTGCCTACTGCCTCGGA	1.660725749	Prdm1.3	0.278969949
Prdm1.4	TGCTGTTGACAGTGAGCGCGGCCATTCTTTGTAGATAATTTAGTGAAGCCACAGATGTAAATTATCTACAAAGAATGGCCATGCCTACTGCCTCGGA	2.006744147	Prdm1.4	-2.437468862
Runx3.1	TGCTGTTGACAGTGAGCGCGCCCGAGACTACAAGAAGAAATAGTGAAGCCACAGATGTATTTCTTCTTGTAGTCTCGGGCTTGCCTACTGCCTCGGA	-3.358108943	Runx3	-1.472470526
Runx3.2	TGCTGTTGACAGTGAGCGACAGGTTCAACGACCTTCGATTTAGTGAAGCCACAGATGTAAATCGAAGGTCGTTGAACCTGGTGCCTACTGCCTCGGA	-1.217911265	Runx3.2	-1.27247758
Runx3.3	TGCTGTTGACAGTGAGCGAGACAACAGCATCTTTGACTCCTATAGTGAAGCCACAGATGTATAGGAGTCAAAGATGCTGTTTGCCTACTGCCTCGGA	ND	Runx3.3	-1.613589553
Rxra.2	TGCTGTTGACAGTGAGCGCGCAGTCTCAGGATTATAGAAATAGTGAAGCCACAGATGTATTTCTATAATCCTGAGACTGCTTGCCTACTGCCTCGGA	0.547686797	Rxra.2	-5.025614668
Rxra.3	TGCTGTTGACAGTGAGCGCGGACATGCCTGTAGAGAAGATTAGTGAAGCCACAGATGTAATCTTCTCTACAGGCATGTCCTTGCCTACTGCCTCGGA	2.780416342	Rxra.3	0.69903302
Rxra.4	TGCTGTTGACAGTGAGCGACGCTCCTCAGGCAAACACTATTAGTGAAGCCACAGATGTAATAGTGTTTGCCTGAGGAGCGGTGCCTACTGCCTCGGA	-0.886518612	Rxra.4	0.518822173
Sgk1.1	TGCTGTTGACAGTGAGCGCCGGCTGAGATGTACGACAATATAGTGAAGCCACAGATGTATATTGTCGTACATCTCAGCCGTTGCCTACTGCCTCGGA	0.893064668	Sgk1	-1.813408833
Sgk1.2	TGCTGTTGACAGTGAGCGCCACGGCTGAGATGTACGACAATAGTGAAGCCACAGATGTATTGTCGTACATCTCAGCCGTGTTGCCTACTGCCTCGGA	3.539408657	Sgk1.2	0.46655946
Smarca1.1	TGCTGTTGACAGTGAGCGCGCAACGAGAATGGTATACAAATAGTGAAGCCACAGATGTATTTGTATACCATTCTCGTTGCATGCCTACTGCCTCGGA	-1.313977133	Smarca1	-3.488886728
Smarca1.2	TGCTGTTGACAGTGAGCGAGGAGAGTCATCTCTAAGAAATTAGTGAAGCCACAGATGTAATTTCTTAGAGATGACTCTCCCTGCCTACTGCCTCGGA	0.419453981	Smarca1.2	-3.697123453
Smarca1.3	TGCTGTTGACAGTGAGCGCGCTGATGTGGTCATCCTATATTAGTGAAGCCACAGATGTAATATAGGATGACCACATCAGCATGCCTACTGCCTCGGA	1.10885531	Smarca1.3	-1.897692495
Smarca4	TGCTGTTGACAGTGAGCGCCCTGGCTCTGAGTATTTCATTTAGTGAAGCCACAGATGTAAATGAAATACTCAGAGCCAGGTTGCCTACTGCCTCGGA	-0.208598847	Smarca4	2.130349101
Smarca4.2	TGCTGTTGACAGTGAGCGAGACAGGAGTACCTCAACAGCATATAGTGAAGCCACAGATGTATATGCTGTTGAGGTACTCCTTGCCTACTGCCTCGGA	ND	Smarca4.2	11.29261534
Smarca4.3	TGCTGTTGACAGTGAGCGAGCGAGCAAGAAGCAGAAGAAACATAGTGAAGCCACAGATGTATGTTTCTTCTGCTTCTTGCTTGCCTACTGCCTCGGA	ND	Smarca4.3	-2.098728125
Tbx21	TGCTGTTGACAGTGAGCGACCAGGAAGTTTCATTTGGGAATAGTGAAGCCACAGATGTATTCCCAAATGAAACTTCCTGGCTGCCTACTGCCTCGGA	1.499785092	Tbx21	-1.238673193
Tbx21.2	TGCTGTTGACAGTGAGCGAGCAGACAGAGATGATCATCACATAGTGAAGCCACAGATGTATGTGATGATCATCTCTGTCTGTGCCTTGCCTCGGA	ND	Tbx21.2	6.243033436
Tbx21.3	TGCTGTTGACAGTGAGCGAGCACACACGTCTTTACTTTCCATAGTGAAGCCACAGATGTATGGAAAGTAAAGACGTGTGTGCCTACTGCCTCGGA	ND	Tbx21.3	5.537532186

Cdk9.1	TGCTGTTGACAGTGAGCGCCAAGATTCTACAGCTCCTAAATAGTGAAGCCACAGATGTATTTAGGAGCTGTAGAATCTTGATGCCTACTGCCTCGGA
Cdk9.2	TGCTGTTGACAGTGAGCGAGAAGGTGAAGGACCGTCTGAATAGTGAAGCCACAGATGTATTCAGACGGTCCTTCACCTTCCTGCCTACTGCCTCGGA
Cdk9.3	TGCTGTTGACAGTGAGCGATGAACCTAATTGAGATTTGTATAGTGAAGCCACAGATGTATACAAATCTCAATTAGGTTCACTGCCTACTGCCTCGGA

Supplemental Experimental Procedures

Large-scale sub-cloning of shRNAmirs into retroviral vectors. To generate the "TF-30" mini-library we subcloned shRNAmir inserts from 96 clones derived from commercially available libraries (Thermo Fisher Scientific, Inc. and TransOmic Technologies, Inc.) into our pLMPd-Amt vector (see Results) using pooled strategies. The majority of clones were selected initially from the GIPZ mouse shRNA whole-genome library (Thermo Fisher). The remaining clones were top-scoring shRNAmir designs from shERWOOD analysis (TransOmic Technologies, Inc.), an algorithm trained on data from a high-throughput functional shRNA-sensor assay (Fellmann et al., 2011). These clones are highlighted in blue (**Table S1**). More reliable RNAi was achieved using shERWOOD designed shRNAmirs compared to other commercially available shRNAmirs for a randomly selected subset of genes (M.E.P. and S.C., unpublished data).

We compared two approaches for subcloning all the shRNAmir inserts from a single pool (**Figure S5A and S5B**). In the first approach, purified plasmid DNA from the original library clones was pooled and cleaved with XhoI and EcoRI to release all 96 inserts simultaneously. In the second approach, each individual shRNAmir insert was amplified in 96 parallel PCR reactions from purified plasmid GIPZ clone DNA using primers specific for common regions in the flanking miR-30 sequences, and that included restriction enzyme sequences (XhoI and EcoRI) compatible with directional cloning (restriction enzymes sequences are underlined: XhoI-5'mir30, CAGAAGG<u>CTCGAG</u>AAGGTATATTGCTGTTGACAGTGAGCG; EcoRI-3'mir30, CTAAAGTAGCCCCTT<u>GAATTC</u>CGAGGCAGTAGGCA). All shRNAmir were amplified in 20 μl reactions containing 5 ng of plasmid DNA, 0.4 U of Phusion Hot Start II High-Fidelity DNA polymerase (Thermo Scientific), 1x Phusion HF buffer, 0.5 μM of each primer and 0.2 □M of each dNTPs using the following cycling parameters: 94 °C for 5 min; 16 cycles of 94 °C for 30 s, 54 °C for 30 s and 75 °C for 30 s; 75 °C for 2 min.; amplification of each shRNAmir was confirmed by agarose gel electrophoresis, and the PCR products were pooled and digested with DpnI to destroy remaining template plasmid DNA, and then digested again with EcoRI and XhoI to generate sticky ends (**Figure S6**). In both approaches, the inserts were gel purified after restriction enzyme digestion using Qiaquick columns. Inserts prepared with each method were ligated separately into 100 ng of pLMPd-Amt in a 2.5:1 molar ratio (Rapid Ligation Kit, Thermo Scientific), and 10% of each ligation was transformed into 50 μl of XL10-Gold chemically competent cells (Agilent Technologies). 384 colonies from each transformation were picked and cultured in 96-well plates to prepare glycerol stocks, which were sent for standard DNA sequencing (Beckman Genomics, Inc.). The BLAST tool was used align all reads to a reference file built from the 96 shRNA sequences we had selected to clone from the original GIPZ library. Each cloning approach resulted in recovery of ~85% of intended shRNAmirs with similar fidelity after sequencing 384 clones, and 100% of shRNAmirs were recovered as perfect matches when clones from both cloning approaches were considered together (data not shown). All constructs are freely available upon request.

CD8$^+$ T cell isolation and stimulation. P14 CD8$^+$ T cells were isolated from whole splenocytes by negative selection using magnetic beads (Invitrogen) and resuspended in D-10 (DMEM + 10% FCS (fetal calf serum), supplemented with 2 mM GlutaMAX (Gibco), 100 U/mL Penicillin/Streptomycin (Gibco), and 50 □M BME). 2x10^5 naïve CD8$^+$ T cells were stimulated with anti-CD3 (clone 2C11) and anti-CD28 (clone 37.51) (1 µg/ml) in 96-well plates pre-coated with 100 µg/mL goat anti-hamster IgG (H+L) secondary antibody (Pierce).

CD8 T cell transduction and adoptive transfer. shRNAmir-RVs were packaged in a 96-well plate format. MicroRNA-adapted short hairpin RNA expressing retroviral vectors (pLMPd-Amt) were used to generated retrovirus from the PLAT-E cell line. PLAT-E cells were seeded in a 96-well plate (6x10^4 cells/well) 1 day before the transfection. The following morning, each well was transfected with 0.2 µg of shRNAmir vector and 0.2 µg of pCL-Eco using TransIT-LT1 (Mirus). Each well of PLAT-E was transfected with a different shRNAmiR construct. Thus, for the shRNAmir library described in this manuscript, the 96 constructs were transfected individually in 96 wells of PLAT-E (**Figure 1B**). 36 and 48 hours after the transfection, 70 µl of the RV-containing supernatant was harvested, stored at 4°C and replaced with fresh medium. 60 hours after transfection, supernatant was harvested for a third time, combined with the stored supernatant and used for transductions. RV supernatant from an individual well was used to transduce CD8 T cells. 96 independent RV supernatants were done in parallel (**Figure 1B**). This was done to ensure that CD8 T cells were transduced with a single shRNAmir construct. Plates containing stimulated CD8 T cells were spun down, and the medium was removed and replaced with RV supernatant supplemented with 50 □M BME and 8 µg/mL polybrene (Millipore) followed by spinfection (60 min. centrifugation at 2000 rpm and 37°C). After transduction, the RV-containing medium was replaced with D-10. 24 hours after the transduction, cells were pooled and 5x10^5 P14 cells were transferred into B6 mice (**Figure 1B**). The remaining cells were sorted based on the top 20% Ametrine1.1 MFI of Amt$^+$ cells as the "Input" sample (**Figure 1B**), DNA was isolated using the FlexiGene DNA kit (Qiagen). For analysis of target gene knockdown, naïve P14 cells were activated in vitro and infected with concentrated shRNAmir-RV supernatants as described (Pipkin et al., 2010). 24 hours after infection, transduced cells were sorted to >95% purity by FACS and cultured until day 6 in media containing either 10 or 100 U/mL rhIL-2 exactly as described (Pipkin et al., 2010).

In vivo CD8 T cell procedures. Mice were infected with 1.5 x 10^5 plaque forming units (PFU) of LCMV clone 13, within 1 hour after cell transfer. 7 days after infection, spleens from 8 recipient mice were combined and CD8 T cells from whole splenocytes were isolated by positive selection using CD90.1 magnetic beads (Miltenyi) and donor P14 CD8 were sorted based on the top 20% Ametrine1.1 MFI of Amt$^+$ cells using a BS FACSAria (BD Biosciences), then based on KLRG1 and CD127 expression, top 20% of Amet+ cells were sorted into effector and memory precursor populations with effector cells defined as KLRG1^{hi}CD127^{lo} and memory precursor as CD127^{hi}KLRG1^{lo} cells (**Figure 1B**). 8x10^5 KLRG1^{hi}CD127^{lo} effector cells and 1.1x10^5 CD127^{hi}KLRG1^{lo} memory precursor cells were routinely obtained after sort. DNA was isolated from both populations using the FlexiGene kit (Qiagen).

CD4 T cell isolation and stimulation. SMARTA CD45.1$^+$ CD4 T cells were isolated from whole splenocytes by negative selection using magnetic beads (Miltenyi) and resuspended in D-10 (DMEM + 10% FCS (fetal calf serum), supplemented with 2 mM GlutaMAX (Gibco) and 100 U/mL Penicillin/Streptomycin (Gibco)) with 50 μM BME and 2 ng/ml mIL-7 (PeproTech). 2x10^5 naïve CD4 T cells per well were stimulated in 96-well plates pre-coated with 8 μg/mL anti-CD3 (clone 2C11, BioXcell) and anti-CD28 (clone 37.51, BioXcell). Cells were plated in the center of two 96-well plates (48 wells in each plate) to avoid the edges of the plates. Empty wells were filled with D-10 medium.

CD4 T cell transduction and adoptive transfer. First, shRNAmir-RVs were packaged in a 96-well plate format. MicroRNA-adapted short hairpin RNA expressing retroviral vectors (pLMD-Amt) were used to generated retrovirus from the PLAT-E cell line. PLAT-E cells were seeded in a 96-well plate (4x10^4 cells/well) 1 day before the transfection. The following morning, each well of PLAT-E was transfected with 0.2 μg of shRNAmir vector and 0.1 μg of pCL-Eco using polyethylenimine (Sigma-Aldrich). Each well of PLAT-E was transfected with a different shRNAmir construct. Thus, for the shRNAmir library described in this manuscript, the 96 constructs were transfected individually in 96 wells of PLAT-E (**Figure 3A**). 24 hours after the transfection, the RV-containing supernatant was harvested, stored at 4°C and replaced with fresh medium. 48 hours after transfection, supernatant was harvested for a second time, combined with the stored supernatant and used for transductions.

RV supernatant from an individual well was used to transduce a single well of CD4 T cells. 96 independent RV supernatants were done in parallel (**Figure 3A**). This was done to ensure that CD4 T cells were transduced with a single shRNAmir construct. The plates containing the stimulated CD4 T cells were spun down and the medium was removed and replaced with RV supernatant from the corresponding well, supplemented with 50 □M BME and 8 μg/mL polybrene (Millipore), followed with centrifugation for 90 min at 1500 rpm and 37°C. CD4 T cells were transduced twice: a first time at 24 hours after in vitro stimulation of CD4 T cells and the second time 12 hours later. After each transduction, the RV-containing medium was replaced with D-10 + 50 mM BME + 10 ng/ml hIL-2. 24 hours after the second transduction, cells were transferred into 48-well plates in fresh D-10 + 50 mM BME + 10 ng/ml hIL-2 and cultured for 2.5 additional days (*in vitro* expansion, **Figure 3A**). Medium was replaced with D-10 +50 mM BME + 2 ng/ml hIL-7 for 16 hours prior to cell sorting (*in vitro* rest, **Figure 3A**). After the *in vitro* IL-7 rest, CD4 T cells were harvested and a small aliquot of each well was kept to verify the transduction efficiencies of individual shRNAmir-RV constructs. Remaining cells were combined and Amt^{hi} cells (top 50% MFI of Amt^{+} cells) were sorted from the pooled CD4 T cells. $5x10^{5}$ sorted Amt^{hi} cells were adoptively transferred in B6 mice (5 mice per group). From the remaining cells, DNA was isolated from $5x10^{5}$-$1x10^{6}$ Amt^{hi} cells as the "Input" sample (**Figure 3A**) using the FlexiGene DNA kit (Qiagen).

Mouse infections and isolation of Tfh and Th1 CD4 T cells. Cells were "rested" for 3.5 days in B6 mice prior to infection. This allows sufficient time for the cells to return to a quiescent state(Johnston et al., 2009; McKinstry et al., 2007). Mice were infected 3.5 days after cell transfer with 2 x 10^{6} PFU of LCMV Armstrong. 6 days after infection, spleens from 5 recipient mice were combined and CD4 T cells from whole splenocytes were isolated by negative selection using magnetic beads (Miltenyi) and donor SMARTA CD4 (CD4^{+}CD45.1^{+}Ametrine^{hi}CD44^{hi}CD62L^{low}) were sorted using a BS FACSAria (BD Biosciences) into Tfh and Th1 populations based on CXCR5 and SLAM expression with Tfh cells defined as SLAM^{lo}CXCR5^{high} and Th1 as SLAM^{hi}CXCR5^{-} cells (**Figure 3A**)(Johnston et al., 2009). $1.5x10^{6}$ Th1 and $2.5x10^{6}$ Tfh cells were routinely obtained after sort. DNA was isolated from each population as described above ("Tfh" and "Th1" samples).

Next generation sequencing. The shRNAmir library was sequenced using the Ion Torrent technology from Life Technologies. Deep sequencing libraries were generated by PCR amplification of shRNA guide strands using primers that tag the product with standard Ion Torrent adaptors (the Ion Torrent adaptors are underlined: primerA+mir30, CCATCTCATCCCTGCGTGTCTCCGACTCAG CTCGAGAAGGTATATTGCT; primerP1+Loop, CCTCTCTATGGGCAGTCGGTGAT TACATCTGTGGCTTCACTA). All shRNAmir in the library are amplified in three individual 50µl PCR reaction containing 100ng of genomic DNA, 1U of Phusion Hot Start II High-Fidelity DNA polymerase (Thermo Scientific), 1x Phusion HF buffer, 0.5µM of each primer and 0.2µM of each dNTPs using the following cycling parameters: 98 °C for 2 min; 26 cycles of 98 °C for 30 s, 58 °C for 30 s and 72 °C for 20 s; 72 °C for 2 min. The shRNA library PCR amplicon was then cleaned using Agencourt AMPure XP beads (Beckman Coulter). The quality of the library after cleanup was assessed using the Agilent High Sensitivity DNA Kit and the Agilent 2100 Bioanalyzer (Agilent Technologies). The library was diluted and used to prepare template-positive Ion Sphere Particles (ISPs) with the Ion OneTouch 200 Template Kit v2 and the Ion One Touch instrument (Life Technologies). The template-positive ISPs were enriched with the Ion OneTouch 200 Template Kit v2 and the Ion OneTouch ES (Life Technologies). The enriched template-positive ISPs were loaded onto an Ion 314 Chip, sequenced with the Ion PGM 200 Sequencing Kit and the Personal Genome Machine System (Life Technologies). 250,000-600,000 reads were routinely obtained, per sample. In some experiment the higher density Ion 318 Chips were used.

Sequencing analysis. The sequencing data was first analyzed by using BLAST with default mismatch settings, to align the sequencing reads to a reference library. To generate the reference library, the sequences ACAGTGAGCG and TAGTGAAGCCACAGATGTA were added to the 5' and 3' ends of the shRNA sense strand, respectively, of each clone.

For CD8 T cell experiments: the total number of reads after sequencing shRNAmirs in the different T cell subsets was scaled to the input sample, and the number of reads for each individual shRNAmir was normalized accordingly. To provide a sufficient baseline to detect possible shRNA depletion in differentiated T cell samples, shRNAs with read counts lower than 50 in the Input were excluded from the analysis. Samples with read counts lower than 50 in the differentiated T cell samples were considered for the analysis if their number of reads was higher than 50 in the Input. Some samples were not detected in the differentiated T cell samples and were considered as depleted. Their read counts were set at one for the analysis. To quantify enrichment of shRNAmirs in the CTL subsets, the read number for each shRNAmir in memory precursor CTL was divided by the corresponding shRNAmir read number in effector CTL, and the resulting values were transformed to $\log_2$. The average and standard deviation of the ratios of shRNAmirs targeting negative control genes (*Cd4*, *Cd14*, *Cd19*, *Ms4a1* (CD20)) was determined and these values were used to calculate Z-scores for each shRNAmir and were plotted (**Figure 2A**). In a separate analysis, the normalized effector or memory precursor CTL reads were divided by the Input reads and the $\log_2$ values for all ratios were calculated and plotted against each other (**Figure 2B**).

For CD4 T cell experiments: the number of reads for each shRNAmir was normalized to its number of reads in the Input sample. Normalized Tfh reads were divided by the normalized Th1 reads to obtain Tfh/Th1 ratios. The $\log_2$ of these ratios was calculated. Conventional Z-score values for every shRNA ratio were calculated using the mean and standard deviation of the negative control set of shRNAs (*Cd14,Cd19 Cd22, Ms4a1* (CD20), *Cd8, Smarca1*) (**Figure 3B**). In a separate analysis, the normalized Tfh or Th1 reads were divided by the Input reads and the $\log_2$ values for all ratios were calculated and plotted against each other (**Figure 3C**). To provide a sufficient baseline to detect possible shRNA depletion in differentiated T cell samples, samples with read counts lower than 50 in the Input were excluded from the analysis. Samples with read counts lower than 50 in the differentiated T cell samples were considered for the analysis if their number of reads was higher than 50 in the Input. Some samples were not detected in the differentiated T cell samples and were considered as depleted. Their read counts were set at one for the analysis. A Z score of $\geq |3.0|$ for at least one shRNAmir against a gene of interest was the primary selection criterion for further consideration. If shRNAmirs against a gene gave conflicting outcomes (e.g., Tfh bias for one, Th1 bias for another), the gene was discarded.

Supplemental Detailed Protocols

Detailed CD8 protocol

Materials

- Platinum-E (PLAT-E) cells
- 150-mm tissue culture dishes (Fisher, cat. # 0877224)
- DMEM (Fisher, cat. # MT10017CV)
- Fetal bovine serum (Omega scientific, cat. # FB-12)
- Penicillin/Streptomycin (Life Technologies, cat. # 15140163)
- Glutamax (L-glutamine, Life Technologies, cat. # 35050079)
- HEPES (1M) (Life Technologies, cat. # 15630-080)
- 2-mercaptoethanol (BME)
- 0.25% Trypsin/EDTA (Life Technologies, cat. # 25200114)
- PBS (Fisher, cat. # MT21040CV)
- Bovine serum albumin (Sigma Aldrich, cat. # A7906)
- UltraPure 0.5M EDTA, pH 8.0 (Life Technologies, cat. # 15575-020)
- Sodium azide (Fisher, cat. # BP922I-500)
- TransIT-LT1 (Mirus, Cat.# MIR 2300)
- Opti-MEM® I Reduced Serum Medium (Life Technologies, cat. # 31985-070)
- pCI-Eco (Addgene, cat. #12371)
- Anti-CD3 antibody (Homemade, clone 2C11)
- Anti-CD28 antibody (BD-Pharm, clone 37.51, cat. # 553295)
- Goat Anti-Hamster IgG (H+L) (ThermoScientific, cat. # 31115)
- Recombinant human IL-2 (PeproTech, cat. # 200-02)
- 70-µm cell strainer (Fisher, cat. # 22363548)
- 1 mL SlipTip Sterile Syringe (without needle) (Fisher, cat. # 14823434)
- ACK Lysing buffer (Life Technologies, cat. # A10492-01)
- 1 mL syringe (with 27 G needles) (Fisher, cat. # 14823434)
- Dynal Mouse CD8 Negative Isolation Kit (Invitrogen, cat. # 114.17D)
- CD90.1 MicroBeads (Miltenyi, cat. # 130-094-523)
- QuadroMACS Separator (Miltenyi, cat. # 130-090-976) and MACS MultiStand (Miltenyi, cat. # 130-042-303)
- MACS LS Columns (Miltenyi, cat. # 130-042-401)
- Polybrene (Millipore, cat. # TR-1003-G)
- Tissue culture-treated 96 well flat plate (Fisher, cat. # 087721G)
- Sterile 96 well round bottom plate (Fisher, cat. # 07200760)
- 15 mL conical tube (Fisher, cat. # 1495949B)
- 50 mL conical tube (Fisher, cat. # 1443222)
- CD45.1^{+} P14 TCR transgenic mice (P14 mice can be purchased from Taconic Farms, cat. #4138)
- C57Bl/6 mice (The Jackson Laboratory (Maine), cat. # 000664)
- LCMV clone13
- FlexiGene DNA kit (Qiagen cat. # 51204)
- Anti-KLRG-1 APC (BioLegend, cat. # 138412)

- Anti- IL-7Rα PE (BioLegend, cat. # 135010)
- Anti-CD8 PerCP-Cy5.5 (eBioscience, cat. # 45-0081)
- Anti-CD90.1 PE-Cy7 (BioLegend, cat. # 202518)
- Anti-CD4 APC-eFluor780 (eBioscience, cat. # 47-0042)
- Anti-B220 APC-eFluor780 (eBioscience, cat. # 47-0452)
- Ethanol (Fisher, Cat. # BP28184)
- Isopropanol (Fisher, Cat. # BP26324)

Reagent setup

- Complete medium (PLAT-E medium): 500 mL DMEM + 10% FBS + 10 mM HEPES + 50 U/mL Penicillin/Streptomycin + 2 mM Glutamax+ 50 μM BME
- cTCM (complete T-cell medium): 500 mL DMEM + 10% FBS + 10 mM HEPES + 50 U/mL Penicillin/Streptomycin + 2 mM Glutamax+ 50 μ□□□□□
- MACS buffer: 500 mL PBS + 0.5% bovine serum albumin + 2 mM EDTA
- FACS buffer: 500 mL PBS + 1% Fetal bovine serum

Equipment

- Flow cytometer: BD Biosciences or any that permits use of Ametrine (violet laser, 505LP filter)
- Fluorescence-activated cell sorting (FACS) machine: BD Biosciences FACSAria or any that permits use of Ametrine (violet laser, 505LP filter)

Production of retroviruses (RV) (days 1-2)

1. Plate PLAT-E in 96 well plates in the afternoon. Seed $6\text{x}10^4$ cells per well in 100 μL of complete medium without Penicillin/Streptomycin. Avoid the edges of the plates by plating cells only in the central 60 wells. Fill the remaining wells with 100 μL of medium.
2. Proceed to transfection of PLAT-E on the following afternoon.
3. Change medium 5 hrs before transfection. Remove medium from PLAT-E and carefully replace with 100 μL of pre-warmed (37°C) complete medium without Penicillin/Streptomycin.
4. Warm TransIT-LT1 Reagent and Opti-MEM® I Reduced Serum Medium to room temperature and vortex gently before using.
5. Prepare transfection mix in round-bottom 96 well plates.
6. Add 0.2 μg shRNAmir vector + 0.2 μg pCL-Eco vector to 10 μL Opti-MEM® I Reduced Serum Medium, pipet gently to mix completely.
7. Add 0.6 μL TransIT-LT1 to 10 μL Opti-MEM® I Reduced Serum Medium, then immediately add 10 μL TransIT-LT1/Opti-MEM® I Reduced Serum Medium to the diluted DNA mixture, pipet gently to mix completely, incubate at room temperature for 20 min.
8. Add the TransIT-LT1 Reagent/DNA complexes drop-wise to different areas of the wells. Gently swirl the plate to mix and incubate at 37°C overnight.
9. 16 hours later, remove medium from PLAT-E and carefully replace with 100 μL of pre-warmed (37°C) cTCM.

P14 CD8 T cell stimulation (day 4)

10. Coat 96 well plates with 50 μL of 100 μg/mL Goat Anti-Hamster IgG (H+L) diluted in PBS (avoid edges of the plates as in step 1). Incubate plates at room temperature for 5 hrs.
11. Isolate P14 CD8 T cells from pooled spleens of $CD45.1^{+}$P14 TCR Tg mice (2 spleens are required to obtain enough CD8 T cells for 96 wells) using the Dynal Mouse CD8 Negative Isolation Kit.
    a. Harvest spleen and transfer to 15 mL tube with 2 mL cTCM
    b. Pour up to 2 spleens into 70 μ□ filter on top of a 50 mL tube. Mash spleens through filter with the thumb end of a 1mL syringe plunger.
    c. Rinse once with 8 mL cTCM, mash again, and then rinse with 10 mL cTCM.

d. Harvest cells at 300 x g for 10 min at 4°C.
e. Resuspend cells in 1 mL ice cold 1x PBS/1.0% FBS and transfer to an eppendorf tube.
f. Combine cells from 1 spleen and the depletion antibody cocktail in the following order and ratios:

   1,000 µL cells

   200 µL FBS

   100 µL depletion antibody cocktail
g. Incubate cells, rotating with antibody cocktail for 20 min at 4°C. Pre-wash Dynal beads three times with ice cold PBS/1.0%FBS during incubation [use 1 mL beads per 10^8 input cells]. Resuspend Dynal beads in vial and transfer required volume to an appropriately sized tube.
h. After incubation, transfer cells/depletion antibody mixture to a fresh 15 mL conical, wash cells by adding up to 15 mL with ice cold PBS/1.0%FBS, rinse eppendorf with 1 mL PBS/1.0%FBS and pool. Harvest cells at 300 x g for 10 min at 4°C, aspirate supernatant, resuspend cell pellet in 7 mL PBS/1.0%FBS in a 15 mL conical tube.
i. Add pre-washed magnetic beads to resuspended cells; incubate rotating at room temperature for 15 min.
j. After incubation, keep inverting tube and immediately apply to magnet, leave tube on magnet ~2 min.
k. With tube remaining on magnet, collect the supernatant by pipetting- transfer to a fresh 15 mL conical tube, add 8 mL fresh ice cold cTCM. Invert tube on magnet to rinse beads/tube.
l. Collect the rinsing volume and pool with first fraction of cells.
m. Harvest cells at 300 x g for 10 min at 4°C.
n. Resuspend cells in 1 mL fresh cTCM and transfer to a 4 mL round bottom tube and place this tube on magnet for 2 min to eliminate any residual beads that will be present.
o. Recover the cells and transfer to a fresh tube and rinse tube on magnet 1x with fresh cTCM.
p. Count cells. Adjust cell concentration to $2x10^6$ cells/mL in cTCM.
q. Prepare the activation medium (2 µg/mL of anti-CD3 and anti-CD28 in cTCM).
r. Rinse pre-coated plates twice with room temperature cTCM before plating cells
s. Combine equal volumes of CD8 T cells and the activation medium; mix and immediately dispense 200 µL into each Goat Anti-Hamster IgG (H+L) pre-coated wells. Avoid the edges of the plates as in step 1. Fill in the used wells with warm cTCM.

12. Harvest retrovirus (RV)-containing supernatant from PLAT-E in the morning and afternoon. Pipet 70 µL of the medium from the PLAT-E into fresh 96 well plates. Store the plates containing the RV at 4°C. Gently add 70 µL of fresh complete medium to the PLAT-E.

Transduction (day 5)

13. The optimal time for the first transduction is 18 hours post stimulation.
14. Pre-warm the centrifuge to 37°C.
15. Harvest the RV-containing supernatant as in step 12 and mix with the corresponding supernatant stored at 4°C.
16. Add 2-mercaptoethanol to a final concentration of 50 µM and 8 µg/mL polybrene to RV supernatant and mix.
17. Carefully add RV mixes to the CD8 T cells in the 96 well plates.
18. Spin at 2,000 rpm for 1 hour at 37°C.
19. Transfer 96 well plates to incubator (37°C 10% CO_2) for an additional 4 hrs.
20. Remove 200 µl of viral supernatants from each well and replace with 200 µL of cTCM and culture at 37°C 10% CO_2 until 48 hrs post activation.

Adoptive transfer and LCMV infection (day 6)

21. 48 hrs post activation, pool cells, determine cell concentration and spin down for 5 minutes at 1500 rpm before adoptive transfer.
22. Discard supernatant and resuspend cells with plain DMEM in 2.5×10^6 cells/mL and 200 μL of the cell solution (corresponding to 5×10^5 cells) is injected into C57Bl/6 recipient mice by the retro-orbital route using an insulin needle.
23. To reduce variation in individual mice and increasing the cell number available for the following procedures, cells are injected into 8 recipient mice.
24. Sort the remaining cells based on the top 20% Ametrine1.1 MFI of Amt^+ cells as the “Input” sample. Genomic DNA was isolated from 3×10^5 sorted cells for the “input” library. (refer to steps 34-47).
25. 1 hour after adoptive transfer, infect transferred mice with LCMV-clone13.
26. Prepare a solution of LCMV-clone13 at a concentration of 2×10^5 PFU/mL in plain DMEM.
27. Inject 500 μL of the LCMV-clone13 solution intra-peritoneally per mouse with a 1 mL syringe. Inject 250 μL on each side of the mouse.

Sort effector and memory precursor populations after LCMV infection (day 13)

28. 7 days after LCMV infection, mice are euthanized and spleens harvested. Spleens from 8 recipient mice are pooled at this step.
29. Total CD8 T cells are isolated with CD90.1 MicroBeads from Miltenyi.
    a. Harvest spleen and transfer to 15 mL tube with 2 mL cTCM
    b. Pour spleen into 70 uM filter on top of a 50 mL tube. Mash spleens through filter with the thumb end of a 1 mL syringe plunger.
    c. Rinse once with 8 mL cTCM, mash again, and then rinse with 10 mL cTCM.
    d. Harvest cells at 300 x g for 10 min at 4°C.
    e. Discard supernatant and resuspend pellet with 2 mL ACK Lysing buffer and incubate at room temperature for 3 min.
    f. Add 8 mL cTCM, mix and take an aliquot to determine cell concentration.
    g. Centrifuge cell suspension at 300 x g for 10 min at 4°C. Aspirate supernatant completely.
    h. Resuspend cell pellet in 90 μL of MACS buffer per 10^7 total cells.
    i. Add 10 μL of CD90.1 MicroBeads per 10^7 total cells.
    j. Mix well and incubate for 15 minutes at 4°C.
    k. Wash cells by adding 2 mL of MACS buffer per 10^7 cells and centrifuge at 300 x g for 10 min at 4°C. Aspirate supernatant completely.
    l. Resuspend up to 10^8 cells in 500 μL of MACS buffer.
    m. Place column in the magnetic field of a suitable MACS Separator.
    n. Prepare column by rinsing with 3 mL of MACS buffer.
    o. Apply cell suspension onto the column. Collect flow-through containing unlabeled cells.
    p. Wash column with 3 mL of MACS buffer. Collect unlabeled cells that pass through and combine with the effluent from step o.
    q. Remove column from the separator and place it on 15 mL tube.
    r. Pipette 5 mL of MACS buffer onto the column. Immediately flush out the magnetically labeled cells by firmly pushing the plunger into the column.
30. Isolated CD8 are stained with the following antibody panel prior to sort
    a. PE: IL-7Rα
    b. PerCP-Cy5.5: CD8
    c. PE-Cy7: CD90.1
    d. APC: KLRG-1

e. APC-eFluor780: CD4; B220

31. After staining, wash cells twice with FACS buffer and resuspend cells in FACS buffer at a concentration appropriate for cell sorting.
32. Sort P14 effector and memory precursor populations based on the top 20% Ametrine1.1 MFI of Amt^+ cells
    a. P14 effector: $CD4^-$ $B220^-$ $CD8^+$ $CD90.1^+$ $Ametrine^+$ $KLRG\text{-}1^+$$IL\text{-}7R\alpha^-$
    b. P14 memory precursor: $CD4^-$ $B220^-$ $CD8^+$ $CD90.1^+$ $Ametrine^+$ $KLRG\text{-}1^-$ $IL\text{-}7R\alpha^+$
33. Fill tubes of sorted P14 effector and memory precursor with PBS and spin down 10 minutes at 1500 rpm. Discard supernatant.

Isolation of genomic DNA

34. Genomic DNA is isolated with the FlexiGene DNA kit, following a modified protocol for the isolation of DNA from 1-2x10^6 cultured cells.
35. Add 300 µL Buffer FG1 to the cell pellet and mix by pipetting until the cells are resuspended.
36. Centrifuge for 1 min at 10,000 x g in a fixed-angle rotor.
37. Discard the supernatant and leave the tube inverted on a clean piece of absorbent paper for 2 min, taking care that the pellet remains in the tube.
38. Add 300 µl Buffer FG2/QIAGEN Protease, close the tube, and vortex immediately until the pellet is completely homogenized. Inspect the tube to check that homogenization is complete.
39. Invert the tube 3 times, place it in water bath, and incubate at 65°C for 5 min. Add 300 µl isopropanol (100%) and mix thoroughly by inversion until the DNA precipitate becomes visible as threads or a clump.
40. Centrifuge for 3 min at 10,000 x g.
41. Discard the supernatant and briefly invert the tube onto a clean piece of absorbent paper, taking care that the pellet remains in the tube.
42. Add 300 µl 70% ethanol and vortex for 5 sec.
43. Centrifuge for 3 min at 10,000 x g.
44. Discard the supernatant and leave the tube inverted on a clean piece of absorbent paper for at least 5 min, taking care that the pellet remains in the tube.
45. Air dry the DNA pellet for 5 min until all the liquid has evaporated.
46. Add 200 µl Buffer FG3, vortex for 5 sec at low speed, and dissolve the DNA by incubating for 5 min at 65°C for at least 1 hour.
47. Store the isolated genomic DNA at -20°C until further use.

Detailed CD4 protocol

Materials

- Platinum-E (PLAT-E) cells
- 150-mm tissue culture dishes (Fisher, cat. # 0877224)
- DMEM (Fisher, cat. # MT10017CV)
- Fetal bovine serum (Omega scientific, cat. # FB-12)
- Penicillin/Streptomycin (Life Technologies, cat. # 15140163)
- Glutamax (L-glutamine, Life Technologies, cat. # 35050079)
- 2-mercaptoethanol (BME) (Sigma-Aldrich, cat. # M3148)
- 0.25% Trypsin/EDTA (Life Technologies, cat. # 25200114)
- PBS (Fisher, cat. # MT21040CV)
- Bovine serum albumin (Sigma Aldrich, cat. # A7906)
- UltraPure 0.5 M EDTA, pH 8.0 (Life Technologies, cat. # 15575-020)
- Sodium azide (Fisher, cat. # BP922I-500)
- Polyethylenimine, branched (PEI) (Sigma-Aldrich, cat. # 408727)
- pCL-Eco (Addgene, cat. #12371)
- Anti-CD3e antibody (BioXCell, clone 145-2C11, cat. # BE0001-1)
- Anti-CD28 antibody (BioXCell, clone 37.51, cat. # BE0015-1)
- Recombinant mouse IL-7 (PeproTech, cat. # 217-17)
- Recombinant human IL-2 (PeproTech, cat. # 200-02)
- 70-µm cell strainer (Fisher, cat. # 22363548)
- 1 mL SlipTip Sterile Syringe (without needle) (Fisher, cat. # 14823434)
- ACK Lysing buffer (Life Technologies, cat. # A10492-01)
- 1 mL syringe (with 27 G needles) (Fisher, cat. # 14823434)
- CD4 isolation kit II (Miltenyi, cat. # 130-095-248)
- QuadroMACS Separator (Miltenyi, cat. # 130-090-976) and MACS MultiStand (Miltenyi, cat. # 130-042-303)
- MACS LS Columns (Miltenyi, cat. # 130-042-401)
- Polybrene (Millipore, cat. # TR-1003-G)
- Insulin needles (U-100) (Fisher, cat. # 1482679)
- Tissue culture-treated 24 well flat plate (Fisher, cat. # 7200740)
- Tissue culture-treated 96 well flat plate (Fisher, cat. # 087721G)
- Sterile 96 well round bottom plate (Fisher, cat. # 07200760)
- 15 mL conical tube (Fisher, cat. # 1495949B)
- 50 mL conical tube (Fisher, cat. # 1443222)
- 1.7 mL Posi-Click Eppendorf Tube (Denville, cat. #C2170)
- SMARTA mice (LCMV gp66-77-IA^b specific)
- B6.SJL mice (The Jackson Laboratories (Maine), cat #002014)
- C57Bl/6 mice (The Jackson Laboratory (Maine), cat. # 000664)
- LCMV Armstrong
- FlexiGene DNA kit (Qiagen cat. # 51204)
- Anti-CD45.1 FITC (eBioscience, cat. # 11-0453)
- Anti-SLAM PE (BioLegend, cat. # 115903)
- Anti-CD8 PerCP-Cy5.5 (eBioscience, cat. # 45-0081)

- Anti-B220 PerCP-Cy5.5 (eBioscience, cat. # 45-0452)
- Anti-CD62L PE-Cy7 (eBioscience, cat. # 25-0621)
- Anti-CD44 eFluor450 (eBioscience, cat. # 48-0441)
- Anti-CD4 APC-eFluor780 (eBioscience, cat. # 47-0042)
- Rat anti-mouse CXCR5 (BD Biosciences, cat. # 551961)
- Biotin conjugated AffiniPure Goat anti-rat (H+L) (Jackson Immunoresearch, cat. # 112-065-167)
- Streptavidin APC (eBioscience, cat. # 17-4317)
- Vα2 □□□□ (eBioscience, cat. # 17-5812)
- CD4 PE (eBioscience, cat. # 12-0041)
- Normal mouse serum (Sigma-Aldrich, cat. # M5905)
- Ethanol (Sigma-Aldrich, cat. # E7023)

Reagent setup

- D10: 500 mL DMEM + 10% FBS (50 mL) + 5 mL Penicillin/Streptomycin + 5 mL Glutamax
- MACS buffer: 500 mL PBS + 0.5% bovine serum albumin + 2 mM EDTA
- FACS buffer: 500 mL PBS + 0.5% bovine serum albumin + 0.1% sodium azide
- Breed SMARTA mice with B6.SJL mice to obtain SMARTA CD45.1$^{+/+}$ mice

Equipment

- Flow cytometer: LSRII from BD Biosciences or any that permits use of Ametrine (violet laser, 505LP dichroic filter)
- Fluorescence-activated cell sorting (FACS) machine: BD Biosciences FACSAria or any that permits use of Ametrine (violet laser, 505LP dichroic filter)

Production of retroviruses (RV) (days 1-2)

1. Plate PLAT-E in 96 well plates in the afternoon (~4-5 pm). Seed $4x10^4$ cells per well in 200 µL of D10. Avoid the edges of the plates by plating cells only in the central 60 wells. Fill the remaining wells with 200 µL of medium.
2. Proceed to transfection of PLAT-E on the following morning (~8-9 am).
3. Change medium 1 hour before transfection. Remove medium from PLAT-E and carefully replace with 100 µL of pre-warmed (37°C) D10. Pre-warm plain DMEM to room temperature for Step 5.
4. 30 minutes later, prepare transfection mix in round-bottom 96 well plates.
5. Mix 0.5 µL PEI with 10 µL plain DMEM (pre-warmed to RT), incubate at RT for 10 minutes.
6. Add 0.1 µg shRNAmir vector + 0.2 µg pCL-Eco vector to PEI/DMEM solution, mix gently and incubate 15 minutes at RT. NOTE: It is more convenient to prepare a plate of the shRNAmir vectors with all wells having the same DNA concentration.
7. Gently add DNA/PEI/DMEM solution to the wells, drop by drop. Gently swirl the plate to mix and incubate at 37°C for 6 hours. NOTE: You can work with a multi-channel pipette for steps 5-7.
8. 6 hours later, gently add 100 µL of fresh warm D10 to every well by pipetting down on the sides of the wells.

SMARTA CD4 T cell stimulation (day 3)

9. In the afternoon of the following day (~2-3 pm), isolate CD4 T cells and plate in anti-CD3/CD28 coated wells
10. Coat 96 well plates with 100 µL 8 µg/mL anti-CD3e and anti-CD28 diluted in PBS (avoid edges of the plates as in step 1 by keeping the same plate layout used for the PLAT-E). Incubate plates at 37°C for at least 1 hour.
11. Isolate CD45.1^{+}SMARTA CD4 T cells from pooled spleens of SMARTA CD45.1$^{+/+}$ mice (2 spleens are required to obtain enough CD4 T cells for 96 wells) using the Miltenyi CD4 T cell isolation kit II.

a. Crush the spleens on a 70-µm cell strainer over a 50 mL conical tube with the plunger from a 1 mL sterile syringe. Rinse the strainer twice with 10-13 mL D10.
b. Spin down cells 5 min at 1600 rpm and 4°C.
c. Discard supernatant and resuspend pellet with 4 mL ACK Lysing buffer (2 mL/spleen) and incubate at room temperature for 2 min.
d. Add 16 mL D10 (8 mL/spleen), mix and take an aliquot to determine cell concentration.
e. Spin down cells as in step 11b.
f. Discard supernatant. Resuspend pellet in 40 µL MACS buffer per 10^7 cells. Filter the cells through a 70-µm cell strainer into a new 50 mL conical tube. Add 10 µL antibody cocktail per 10^7 cells. Incubate 10 min at 4°C, shaking every 5 minutes.
    i. During the incubation, prepare D10 + 50 µM BME + 2 ng/ml mIL-7 and pre-warm at 37°C. Calculate 200 µL per well to plate.
g. Add 30 µL MACS buffer per 10^7 cells. Add 20 µL biotin beads per 10^7 cells. Incubate 15 min at 4°C, shaking every 5 minutes.
h. Prepare MACS LS columns
    i. Determine the number of columns to be used. Each column can be loaded with up to $1x10^8$ magnetically labeled cells. Assume that ~75% of the cells from a spleen of a SMARTA TCR Tg mouse are labeled with the CD4 T cell isolation kit II.
    ii. Place LS Column on QuadroMacs Separator placed on the MACS MultiStand. Place a 15 mL conical tube under the column to collect the flow-through.
    iii. Wash column with 6 mL MACS buffer. Prepare the column at the last minute to keep it cold and prevent it from drying.
i. Add 1-2 mL MACS buffer per 10^7 cells and spin down as in step 11b.
j. Resuspend cells in MACS buffer: up to 10^8 cells in 500 µL.
k. Load cells unto column and collect flow-through.
l. Wash column with 5 mL MACS buffer and collect flow-through.
m. Take an aliquot to verify purity of isolated CD4
    i. Stain with the following antibodies
        1. FITC: CD45.1
        2. PE: CD4
        3. PerCP-Cy5.5: CD8, B220
        4. eFluor450: CD44
        5. APC: Vα2
    ii. The isolated cells should be $CD8^-$ $B220^-$ $CD4^+$ $CD45.1^+$ $V\alpha2^+$ and mostly $CD44^-$ (~85-90%).
n. Spin down cells as in step 11b.
o. Resuspend cells in 1 mL pre-warmed D10 + 50 µM BME + 2 ng/mL mIL-7. Take an aliquot and determine cell concentration.
p. Adjust cell concentration to $1x10^6$ cells/mL in D10 + 50 µM BME + 2 ng/mL mIL-7.

12. Remove PBS from the wells of the 96 well plates by flicking.
13. Seed $2x10^5$ SMARTA cells per well by pipetting 200 µL in each well coated with anti-CD3e and anti-CD28 antibodies. Use a multichannel pipette to work faster and to avoid the wells from drying. Fill in the unused wells with warm D10. It is critical to gently pipette the cells in the wells to obtain a uniform distribution of the cells resulting in efficient activation.
14. Harvest retrovirus (RV)-containing supernatants from PLAT-E. Pipet 170 µL of medium from the PLAT-E into fresh 96 well plates for storage at 4°C. Gently add 180 µL of warm fresh D10 to the PLAT-E by pipetting down on the sides of the wells.

1st transduction (day 4)

15. The optimal time for the first transduction is 22-24 hours post stimulation to time with the mitotic phase (~3-4 pm)
16. Pre-warm the centrifuge to 37°C.
17. Harvest the RV-containing supernatants as in step 14 and mix with the corresponding supernatants stored at 4°C.
18. Transfer 150 µL of the RV supernatants to fresh 96 well plates to prepare the RV transduction mixes. Keep the remaining RV supernatant at 4°C for the second transduction.
19. Supplement RV supernatants with 50 µM BME and 8 µg/ml polybrene and mix by pipetting up and down.
20. Spin down the plates of SMARTA CD4 for 1 min at 1500 rpm. Remove medium.
21. Carefully add RV mixes to the CD4 on the 96 well plates by pipetting down on the sides of the wells.
22. Spin for 2 hours at 1500 rpm and 37°C.
23. Prepare D10 + 50 µM BME + 10 ng/mL hIL-2 and pre-warm at 37°C. Calculate 200 µL□□□□□□□□□□
24. Remove supernatant and gently replace with pre-warmed D10 + 50 µM BME + 10 ng/mL recombinant hIL-2.
25. Verify transfection efficiency of all wells of PLAT-E by flow cytometry.
    a. Add 100 µL of FACS buffer to all wells and pipet each well separately into a tube for acquisition on a flow cytometer.

2nd transduction (day 5)

26. Proceed with the second transduction approximately 36 hours post-stimulation (~8 am).
27. Repeat steps 16, 18-25.

IL-2 expansion (day 6)

28. Separately move transduced SMARTA to 24 well plates for expansion 24 hours after the second transduction (~9-10 am).
29. Pre-warm D10 + 50 µM BME + 10 ng/mL hIL-2. Calculate 1 mL per well.
30. Pipet CD4 from each well of the 96 well plates to individual wells of 24 well plates (keep CD4 T cells separated).
31. Add 1 mL of warm fresh D10 + 50 µM BME + 10 ng/mL hIL-2 to each well.

IL-7 *in vitro* rest (day 8)

32. Cells are given fresh D10 + 50 µM BME + 2 ng/mL mIL-7 16-18 hours prior to transfer (~5-6 pm).
33. Pre-warm D10 + 50 µM BME + 2 ng/mL mIL-7. Calculate 1 mL per well.
34. Spin down plates of SMARTA CD4 for 1 min at 1500 rpm. Carefully remove medium.
35. Add 1 mL of warm D10 + 50 µM BME + 2 ng/mL mIL-7.

Sort and adoptive transfer (day 9)

36. Cells are pooled before sorting the transduced cells for adoptive transfer (~8-9 am).
37. Harvest a small aliquot (~25 µL) of cells from each well to verify the transduction efficiency of every shRNAmir construct.
38. Harvest and pool SMARTA from all the wells and verify the transduction efficiency of the pooled cells.
39. Sort cells expressing high levels of Ametrine (top 50%).
40. After sort, the cells are prepared for adoptive transfer. The tube containing the Ametrine-high cells is filled with PBS and the cells are spun down for 10 minutes at 1500 rpm at 4.
41. Discard supernatant and wash cells with plain DMEM; spin cells as in step 40.
42. Discard supernatant and resuspend cells in 1 mL plain DMEM. Determine cell concentration.

43. Cells are diluted to $5x10^6$ cells/mL in plain DMEM and 100 µL of the cell solution (corresponding to $5x10^5$ cells) is injected into C57Bl/6 recipient mice by the retro-orbital route using an insulin needle.
44. To reduce variation caused by variable engraftment in individual mice and to increase the cell number available for the following procedures, cells are injected into 5 recipient mice.
45. Cells are allowed to rest *in vivo* for at least 3.5 days before infection.
46. Genomic DNA is isolated from $1x10^6$ sorted cells for the "input" library. (refer to steps 56-67).

LCMV infection (day 13)

47. Mice are infected with LCMV-Armstrong (~8-9 am).
48. Prepare a solution of LCMV-Armstrong at a concentration of $4x10^6$ PFU/mL in plain DMEM.
49. Inject 500 µL of the LCMV-Armstrong solution (corresponding to $2x10^6$ PFU) intra-peritoneally per mouse using a 1 mL syringe (with 27 G needle). Inject 250 µL on each side of the mouse.

Sort Tfh and Th1 populations after LCMV infection (day 19)

50. 6 days after LCMV infection, mice are euthanized (~8 am) and spleens harvested. Spleens from 5 recipient mice are pooled at this step.
51. Total CD4 T cells are isolated with a CD4 isolation kit II from Miltenyi as described in step 11.
52. Isolated CD4 are stained with the following antibody panel prior to sort (stain cells at a concentration of $2.5x10^7$ cells/mL)
    b. FITC: CD45.1
    c. PE: SLAM
    d. PerCP-Cy5.5: CD8, B220
    e. PE-Cy7: CD62L
    f. eFluor450: CD44
    g. Ametrine
    h. APC: CXCR5 (3 steps staining protocol, detailed below)
    i. APC-eFluor780: CD4
53. CXCR5 3 steps staining protocol
    a. Resuspend isolated CD4 at a concentration of $1.5x10^7$ cells/mL in FACS buffer + serum (2% normal mouse serum + 2% FBS). Add 10 µL/mL (1:100 dilution) of primary CXCR5 antibody (rat anti-mouse CXCR5 from BD). Incubate 1 hour at 4°C, shaking tube every 15 minutes.
    b. Wash twice with FACS buffer by filling the tube with FACS buffer and spinning down 5 min and 4°C.
    c. Resuspend isolated CD4 at a concentration of $1.5x10^7$ cells/mL in FACS buffer + serum (2% normal mouse serum + 2% FBS). Add 1 µL/mL (1:1000 dilution) of secondary antibody (Jackson Immunoresearch biotin conjugated AffiniPure Goat anti-rat (H+L)). Incubate 30 minutes at 4°C, shaking tube every 15 minutes.
    d. Wash twice with FACS buffer as in step 53b.
    e. Resuspend isolated CD4 at a concentration of $1.5x10^7$ cells/mL in FACS buffer. Stain with streptavidin-APC (1:1000) and antibody cocktail detailed in step 52 for 30 minutes at 4°C, shaking tube every 15 minutes.
    f. Wash twice with FACS buffer as in step 53b and resuspend cells in FACS buffer at a concentration appropriate for cell sorting ($1\text{-}2x10^7$ cells/mL).
54. Sort SMARTA Tfh and Th1 populations
    a. SMARTA Tfh: $CD8^-$ $B220^-$ $CD4^+$ $CD45.1^+$ $Ametrine^+$ $CD44^+$ $CD62L^-$ $CXCR5^+$ $SLAM^{lo}$
    b. SMARTA Th1: $CD8^-$ $B220^-$ $CD4^+$ $CD45.1^+$ $Ametrine^+$ $CD44^+$ $CD62L^-$ $SLAM^{high}$ $CXCR5^-$
55. Fill tubes of sorted SMARTA Tfh and SMARTA Th1 with PBS and spin down 10 minutes at 1500 rpm at 4°C. Discard supernatant.

Isolation of genomic DNA

56. Genomic DNA is isolated with the FlexiGene DNA kit from Qiagen following a modified protocol for the isolation of DNA from 1-2x10^6 cultured cells.
57. Add 300 µL Buffer FG1 to the cell pellet and mix by pipetting until the cells are resuspended. Transfer to a 1.7 mL Eppendorf tube.
58. Prepare Buffer FG2 + Qiagen protease that has been previously resuspended in buffer FG3 as per kit's instructions.
59. Add 300 µL Buffer FG2 + 3 µL Qiagen protease, close the tube and invert 3 times. Incubate at 65°C for 10 minutes.
60. Add 600 µL isopropanol (100%) and mix thoroughly by inversion until the DNA precipitate becomes visible.
61. Centrifuge for 10 minutes at top speed in a table top centrifuge. Discard the supernatant and briefly invert the tube onto a piece of clean absorbent paper.
62. Add 600 µL 70% ethanol and vortex for 5 seconds.
63. Centrifuge for 5 minutes at top speed.
64. Discard the supernatant and leave the tube inverted on a clean piece of absorbent paper for 5 minutes.
65. Air-dry the pellet for 5 minutes.
66. Add 300 µL Buffer FG3, vortex for 5 seconds and dissolve the pellet by incubating at 65°C for at least 1 hour.
67. Store the isolated genomic DNA at -20°C until further use.

Detailed CD8 deep sequencing and analysis

Materials

- Qubit dsDNA HS Assay Kit (Life Technologies, cat. # Q32854)
- Qubit Fluorometer (Life Technologies)
- PCR machine
- Agilent 2100 Bioanalyzer (Agilent Technologies)
- High Sensitivity DNA analysis kit (Agilent Technologies, cat. #5067-4626)
- Deep sequencing machine
  - We use the Ion OneTouch 2 System from Life Technologies to prepare the sequencing libraries with the following kits:
    - Ion OneTouch 200 Template Kit v2 DL (Life Technologies, cat. #4480285)
    - Ion Sphere Quality Control Kit (Life Technologies, cat. #4468656)
  - We use the Ion PGM System from Life Technologies to sequence the libraries with the following kits:
    - Ion PGM Sequencing 200 Kit v2 (Life Technologies, cat. #4482006)
    - Ion 314 Chip Kit v2 for up to 4.0-5.5x10^5 reads per sequencing reaction (Life Technologies, cat. #4482261)
    - Ion 318 Chip Kit v2 for up to 4.0-5.5x10^6 reads per sequencing reaction (Life Technologies, cat. #4484355)
- Primers with Ion Torrent adaptors (underlined)
  - A miR30 for: CCATCTCATCCCTGCGTGTCTCCGACTCAG CTCGAGAAGGTATATTGCT
  - Loop rev: CCTCTCTATGGGCAGTCGGTGAT TACATCTGTGGCTTCACTA
- Phusion High-Fidelity DNA Polymerase (2 U/µL) (Thermo Scientific, cat. #F-530)
- Deoxynucleotide (dNTP) Solution Mix (Promega, cat. #U1330)
- Agencourt AMPure XP – PCR Purification (Beckman Coulter, cat. #A63881)
- DynaMag-Spin/2 Magnet (Life Technologies, cat. #12320D/12321D)
- Ethanol (Sigma-Aldrich, cat. # E7023)
- 1.7 mL Posi-Click Eppendorf Tube (Denville, cat. #C2170)
- PCR Tubes, 120 strips with individual lids (Denville, cat. #C18098-3)
- Nuclease-free water (Qiagen, cat. #129115)

Library preparation

1. Determine the DNA concentration of the isolated genomic DNA with a Qubit Fluorometer using the Qubit dsDNA HS Assay Kit.
2. All shRNAmirs in the library were amplified with three individual PCR reactions containing 100 ng of genomic DNA using the following recipe.

5 x Phusion buffer	10 µL
10 mM dNTPs	1 µL
10 □M A miR30 for	□□□ µL
10 □M loop rev	2.5 µL
DNA	100 ng
Phusion Hot Start Taq	0.5 µL

H_2O	Up to 33.5 µL
Total	50 µL

3. Primer sequences with the Ion Torrent adaptors underlined
    a. Primer A+miR30: CCATCTCATCCCTGCGTGTCTCCGACTCAG CTCGAGAAGGTATATTGCT
    b. PrimerP1+Loop: CCTCTCTATGGGCAGTCGGTGAT TACATCTGTGGCTTCACTA
4. Amplification protocol
    a. 98°C – 2 min
    b. 98°C – 30 sec
    c. 58°C – 30 sec
    d. 72°C – 20 sec (b->d 26 cycles)
    e. 72°C – 2 min
5. The PCR reactions are pooled and proceed to the Ampure XP bead cleanup to remove traces of primer dimers.
6. Keep 5 µL of the pooled PCR reaction for step 23.
7. Combine PCR reactions in a lo-bind eppendorf tube for 200 µL final volume (make up volume to 200 µL with nuclease-free H_2O).
8. Add 260 µL Ampure beads per sample.
9. Mix by vortexing carefully.
10. Incubate samples $\geq$ 5 min (important – binding to beads is time dependent)
11. Place sample on magnet and wait $\geq$ 5 min. Aspirate supernatant and keep as backup.
12. Wash beads gently twice with 5 volumes of 70% ethanol, while beads stay on magnet.
13. Remove tubes and quick spin to collect residual ethanol.
14. Replace in magnet and aspirate remaining ethanol.
15. Allow samples to dry until beads appear to crack.
16. Elute DNA by adding 20 µL of Nuclease-free H_2O.
17. Mix by pipetting samples 10 times and incubating at RT for 5 min.
18. Place samples on magnet for 2 min and collect supernatant containing purified DNA.
19. Adjust eluate to 50 µL final with nuclease-free H_2O
20. Repeat the entire procedure (steps 8-18, then skip to 21) using 80 µL Ampure beads.
21. Elute the DNA in 20 µL of nuclease free H_2O.
22. Determine the DNA concentration of the cleaned PCR product as in step 1.
23. Verify that the beads have removed the primer dimers by quantifying the pre-cleaned and cleaned library DNA with the Agilent 2100 Bioanalyzer and the High Sensitivity DNA analysis kit.

PGM sequencing procedure

24. The cleaned library samples are sequenced using the Ion Torrent technology from Life Technologies following the manufacturer's instructions.
25. Dilute the library and use it as a template to prepare template-positive Ion Sphere Particles (ISPs) with the Ion One Touch 200 Template Kit v2 and the Ion One Touch instrument.
26. Perform a quality control test on the template-positive ISPs using the Ion Sphere Quality Control kit and a Qubit Fluorometer. Proceed to enrichment if 10-30% of the ISPs are template-positive.
27. Proceed to the enrichment of the template-positive ISPs with the Ion One Touch 200 Template kit v2 and the Ion One Touch ES.

28. Perform a quality control on the enriched template-positive ISPs using the Ion Sphere Quality Control kit and a Qubit Fluorometer. Proceed to sequencing if more than 50% of the ISPs are template-positive.
29. The enriched template-positive ISPs are loaded on Ion 314 Chip and sequenced with the Ion PGM 200 Sequencing Kit and the Personal Genome Machine System. 2.5-6x10^6 reads are routinely obtained with the Ion 314 Chip.

Analysis of sequencing reads from PGM (CD8)

30. Prepare the reference library for analysis by adding ACAGTGAGCG and TAGTGAAGCCACAGATGTA to the 5' and 3' of shRNA sense strand, respectively, of each clone of the 96 constructs used in the experiment.
31. Align the sequencing data with the reference library using BLAST with default mismatch settings.
32. The number of reads for each shRNAmir in the input, effector and memory precursor samples is used in the analysis.
    a. shRNAmir with read counts lower than 50 in the input were excluded from the analysis.
    b. shRNAmir with read counts lower than 50 in the effector or memory precursor samples were considered for the analysis if their number of reads was higher than 50 in the input.
    c. shRNAmir not detected in the effector or memory precursor samples were considered as depleted and a value of 1 was assigned to these shRNAmir.
33. Normalize the number of reads of every shRNAmir in the effector and memory precursor samples were normalized to their respective number of reads in the input sample.
34. The relative enrichment or depletion of shRNAmir from each population is determined by dividing the normalized reads in the memory precursor sample by the normalized reads in the effector sample to obtain the memory precursor/effector ratios for every shRNAmir in the library.
35. Calculate the $\log_2$ of these ratios.
36. Calculate the Z-score values for every shRNAmir with the following formula:

$$Z\ (shRNAmir\ A) = \frac{Log2\ ratio\ of\ shRNAmir\ A - mean\ of\ log2\ ratios\ of\ ctrl\ shRNAmir}{standard\ deviation\ of\ ctrl\ shRNAmir}$$

    a. The Z-score are calculated using control shRNAmir because several shRNAmir in the library were expected to have an effect and therefore the standard deviation would be large and Z-score values small.
    b. Negative control shRNAmir are specific for genes not expressed in CD8 and therefore are expected to have no effect on the differentiation of effector and memory precursor cells. These shRNAmir were equally distributed in both populations as shown by their clustering the center of the Log_2 and Z-score distributions.
    c. List of negative control shRNAmir used in the CD8 T cell analysis with the current library: *Cd4*, *Cd14*, *Cd19*, *Ms4a1* (CD20).
    d. All shRNAmir can be used to calculate Z-score values when screening a large shRNAmir library. In this case, it would be expected that the majority of shRNAmir would have no effect on effector and memory precursor differentiation.
37. The Z-score values of every shRNAmir can be plotted.
    a. A Z-score of ≥ |3| Z-score for two or more shRNAmirs against a gene of interest is the primary criterion for further consideration.
38. Another method of visualizing the data takes into account the effects shRNAmir have on cell accumulation (possible effects on survival or proliferation).
    a. For this analysis the effector/input and memory precursor/input ratios are obtained by dividing the normalized reads in the effector or memory precursor samples by the reads in the input sample for every shRNAmir in the library.

b. The $\log_2$ values of the effector/input and memory precursor/input ratios are calculated.
c. Plot the $\log_2$ values of the effector/input ratios against those of the memory precursor/input ratios.

Detailed CD4 deep sequencing and analysis

Materials

- Qubit dsDNA HS Assay Kit (Life Technologies, cat. # Q32854)
- Qubit Fluorometer (Life Technologies)
- PCR machine
- Agilent 2100 Bioanalyzer (Agilent Technologies)
- High Sensitivity DNA analysis kit (Agilent Technologies, cat. #5067-4626)
- Deep sequencing machine
  - We use the Ion OneTouch 2 System from Life Technologies to prepare the sequencing libraries with the following kits:
    - Ion OneTouch 200 Template Kit v2 DL (Life Technologies, cat. #4480285)
    - Ion Sphere Quality Control Kit (Life Technologies, cat. #4468656)
  - We use the Ion PGM System from Life Technologies to sequence the libraries with the following kits:
    - Ion PGM Sequencing 200 Kit v2 (Life Technologies, cat. #4482006)
    - Ion 314 Chip Kit v2 for up to 4.0-5.5x10^5 reads per sequencing reaction (Life Technologies, cat. #4482261)
    - Ion 318 Chip Kit v2 for up to 4.0-5.5x10^6 reads per sequencing reaction (Life Technologies, cat. #4484355)
- Primers with Ion Torrent adaptors (underlined)
  - A miR30 for: CCATCTCATCCCTGCGTGTCTCCGACTCAG CTCGAGAAGGTATATTGCT
  - Loop rev: CCTCTCTATGGGCAGTCGGTGAT TACATCTGTGGCTTCACTA
- Phusion High-Fidelity DNA Polymerase (2 U/µL) (Thermo Scientific, cat. #F-530)
- Deoxynucleotide (dNTP) Solution Mix (New England BioLabs, cat. #N0447)
- Agencourt AMPure XP – PCR Purification (Beckman Coulter, cat. #A63881)
- DynaMag-Spin/2 Magnet (Life Technologies, cat. #12320D/12321D)
- Ethanol (Sigma-Aldrich, cat. # E7023)
- 1.7 mL Posi-Click Eppendorf Tube (Denville, cat. #C2170)
- PCR Tubes, 120 strips with individual lids (Denville, cat. #C18098-3)
- Nuclease-free water (Qiagen, cat. #129115)

Library preparation

1. Determine the DNA concentration of the isolated genomic DNA with a Qubit Fluorometer using the Qubit dsDNA HS Assay Kit as per manufacturer's instructions.
2. All shRNAmirs in the library were amplified with three individual PCR reactions containing 100 ng of genomic DNA using the following recipe.

5 x Phusion buffer	10 µL
10 mM dNTPs	1 µL
10 □M A miR30 for	□□□ µL
10 □M loop rev	2.5 µL
DNA	100 ng
Phusion Hot Start Taq	0.5 µL
H_2O	Up to 33.5□µL
Total	50□µL

3. Amplification protocol
   a. 98°C – 2 min
   b. 98°C – 30 sec

    c. 58°C – 30 sec
    d. 72°C – 20 sec -> b-d: 26 cycles
    e. 72°C – 2 min
4. The PCR reactions are pooled and 15 µL is loaded on a 1.5% agarose gel to verify the efficiency of the PCR reaction.
    a. The amplified library should be visible as a band of 127 bp.
5. Proceed to the cleanup of PCR reaction to remove traces of primer dimers.
6. Keep 5 µL of the pooled PCR reaction for step 23.
7. Combine PCR reactions in a 1.7 mL Eppendorf tube for 200 µL final volume (make up volume to 200 µL with nuclease-free H_2O).
8. Add 260 µL of AMPure XP beads per sample.
9. Mix by vortexing carefully.
10. Incubate samples $\geq$ 5 min (important – binding to beads is time dependent)
11. Place sample on magnet and wait $\geq$ 5 min. Aspirate supernatant and keep as backup.
12. Wash beads gently twice with 1 mL of 70% ethanol, while beads stay on magnet.
13. Remove tubes and quickly spin to collect residual ethanol.
14. Replace in magnet and aspirate remaining ethanol.
15. Allow samples to dry until beads appear to crack.
16. Elute DNA by adding 20 µL of nuclease-free H_2O.
17. Mix by pipetting samples 10 times and incubating at RT for 5 min.
18. Place samples on magnet for 2 min and collect supernatant containing purified DNA.
19. Adjust eluate to 50 µL final with nuclease-free H_2O
20. Repeat the entire procedure (steps 46-56, then skip to step 59) using 80 µL of AMPure XP beads.
21. Elute the DNA in 20 µL of nuclease free H_2O.
22. Determine the DNA concentration of the cleaned PCR product as in step 39.
23. Verify that the beads have removed the primer dimers by quantifying the pre-cleaned (kept at step 6) and cleaned library DNA (step 21) with the Agilent 2100 Bioanalyzer and the High Sensitivity DNA analysis kit as per manufacturer's instructions.
    a. The result obtained with the Agilent Bioanalyzer should show the disappearance of the primer dimers after the cleanup of the PCR reaction with the AMPure beads XP.
    b. Only once the primer dimers have been successfully removed from the amplified library that the library can be sequenced.

PGM sequencing procedure

24. The cleaned library samples are sequenced using the Ion Torrent technology from Life Technologies following the manufacturer's instructions.
25. Dilute the library and use it as a template to prepare template-positive Ion Sphere Particles (ISPs) with the Ion One Touch 200 Template Kit v2 DL and the Ion OneTouch 2 System following manufacturer's instructions.
    a. The dilution of the library is critical for an efficient sequencing reaction. The dilution factor to obtain a library that will be efficiently sequenced needs to be experimentally determined. We typically need to dilute the library by a factor of 50 to 267.
    b. The reaction can be started at the end of the day and the template-positive ISPs can be recovered the following morning.
26. Perform a quality control test on the template-positive ISPs using the Ion Sphere Quality Control kit and a Qubit Fluorometer following manufacturer's instructions.
    a. Proceed to enrichment if 10-30% of the ISPs are template-positive.
    b. If less than 10% of the ISPs are template-positive, the sequencing reaction will not generate enough reads for analysis.
    c. If more than 30% of the ISPs are template-positive, some ISPs will be multi-templated and will generate mixed reads.

27. Enrich the template-positive ISPs with the Ion One Touch 200 Template kit v2 DL and the Ion OneTouch System following manufacturer's instructions.
28. Perform a quality control on the enriched template-positive ISPs using the Ion Sphere Quality Control kit and a Qubit Fluorometer following manufacturer's instructions.
    a. Proceed to sequencing if more than 50% of the ISPs are template-positive.
29. The enriched template-positive ISPs are loaded on Ion 314 or Ion 318 Chip and sequenced with the Ion PGM 200 Sequencing Kit and the Personal Genome Machine System following manufacturer's instructions.
    a. 250,000-500,000 reads are routinely obtained with the Ion 314 Chip.
    b. The higher density Ion 318 Chip can yield up to 10 times more reads than the Ion 314 Chip.
30. Prepare the reference library for analysis by adding ACAGTGAGCG and TAGTGAAGCCACAGATGTA to the 5' and 3' of shRNA sense strand, respectively, of each shRNAmir sequences used in the experiment.
31. Align the sequencing data with the reference library using BLAST with default mismatch settings.
32. The number of reads for each shRNAmir in the input, Tfh and Th1 samples is used in the analysis.
    a. shRNAmir with read counts lower than 50 in the input were excluded from the analysis.
    b. shRNAmir with read counts lower than 50 in the Tfh or Th1 samples were considered for the analysis if their number of reads was higher than 50 in the input.
    c. shRNAmir not detected in the Tfh or Th1 samples were considered as depleted and a value of 1 was assigned to these shRNAmir.
33. Normalize the number of reads of every shRNAmir in the Tfh and Th1 samples to their respective number of reads in the input sample.
34. The relative enrichment or depletion of shRNAmir from each population is determined by dividing the normalized reads in the Tfh sample by the normalized reads in the Th1 sample to obtain the Tfh/Th1 ratios for every shRNAmir in the library.
35. Calculate the $\log_2$ of these ratios.
36. Calculate the Z-score values for every shRNAmir with the following formula:

$$Z\ (shRNAmir\ A) = \frac{Log2\ ratio\ of\ shRNAmir\ A - mean\ of\ log2\ ratios\ of\ ctrl\ shRNAmir}{standard\ deviation\ of\ ctrl\ shRNAmir}$$

    a. The Z-score are calculated using control shRNAmir because several shRNAmir in the library were expected to have an effect and therefore the standard deviation would be large and Z-score values small.
    b. Negative control shRNAmir are targeting genes not expressed in CD4 and therefore are expected to have no effect on the differentiation of Tfh and Th1 cells. These shRNAmir were equal ly distributed in both populations as shown by their clustering the center of the Log_2 and Z-score distributions.
    c. List of negative control shRNAmir used in the CD4 T cell analysis with the current library: *Cd14*, *Cd19*, *Cd22*, *Ms4a1* (CD20), *Cd8* and *Smarca1*.
    d. All shRNAmir can be used to calculate Z-score values when screening a large shRNAmir library. In this case, it would be expected that the majority of shRNAmir would have no effect on Tfh and Th1 differentiation.
37. The Z-score values of every shRNAmir can be plotted.
    a. A Z-score of ≥ |3| Z-score for at least one shRNAmir against a gene of interest is the primary criterion for further consideration.
    b. If shRNAmirs against a gene are giving conflicting outcomes (one shRNAmir with a Tfh bias and a Th1 bias for a second shRNAmir), the gene was discarded.
38. Another method of visualizing the data takes into account the effects shRNAmir have on cell accumulation (possible effects on survival or proliferation).
    a. For this analysis the Tfh/input and Th1/input ratios are obtained by dividing the normalized reads in the Tfh or Th1 samples by the reads in the input sample for every shRNAmir in the library.
    b. The $\log_2$ values of the Tfh/input and Th1/input ratios are calculated.
    c. Plot the $\log_2$ values of the Tfh/input ratios against those of the Th1/input ratios.

Detailed shRNAmir sub-cloning protocol

Materials

- GIPZ mouse shRNA whole-genome library (Thermo Scientific)
- Whatman 2 ml deepwell block (Fisher, cat. #05-714-034)
- LB Broth LENNOX (VWR, cat. # EM1.00547.0500)
- Ampicillin (Fisher, cat. # BP1760-25)
- Glycerol (Fisher, cat. # BP229-4)
- Aeraseal (Fisher, cat. # 50212463)
- GenElute™ HP 96 Well Plasmid Miniprep Kit (Sigma Aldrich, cat. # NA9604)
- XhoI (New England Biolabs, cat. # R0146)
- EcoRI (New England Biolabs, cat. # R0101)
- Hot Start II High-Fidelity DNA polymerase (Thermo Scientific, cat. # F-549)
- QIAquick Gel Extraction Kit (Qiagen, cat. # 28706)
- Rapid Ligation Kit (Thermo Scientific, cat. # K1423)
- XL10-Gold chemically competent cells (Agilent Technologies, cat. # 200315)

Culture preparation

1. Transfer 5 µL of glycerol stock of GIPZ clones into a Whatman 2 mL deepwell block containing 1.2 mL of 2X LB with 100 µg/mL Ampicillin.
2. Culture for ~24 hours at 37°C with vigorous shaking (~300 rpm)
3. Transfer 5 µL of the starter cell culture into a 2 mL square well block containing 1.2 mL of 2X LB with 100 µg/mL Ampcillin.
4. Culture for 16-17 hours at 37°C with vigorous shaking (~300 rpm)

96 well DNA miniprep

5. DNA is purified with GenElute™ HP 96 Well Plasmid Miniprep Kit.
6. Pellet the 96 deep well culture plate containing the overnight culture by centrifuging at 1800 x g for 10 minutes.
7. During centrifugation, assemble the manifold by placing an empty deep well collection plate in the manifold base. Place the manifold collar over the deep well collection plate so it fits securely on top of the manifold base. Place the filter plate on top of the manifold collar.
8. Following centrifugation of the deep well culture plate, pour off the supernatant and blot the deep well culture plate upside down on an absorbent pad.
9. Using a multichannel pipette, add 200 µL of Resuspension Solution to the pelleted cells and resuspend by pipetting up and down.
10. Add 200 µL of Lysis Solution and pipette up and down 5-6 times to mix.
11. Allow the lysate to incubate at room temperature for 3 to 5 minutes; do not allow the lysis to exceed 5 minutes.
12. Add 200 µL of Neutralization Solution and pipette up and down 5-6 times to mix.
13. Add 200 µL of Binding Solution and pipette up and down 2-3 times to mix.
14. Transfer the lysate mix from the culture plate to the filter plate that is located on top of the manifold assembly and let sit for 5 minutes.
15. Apply vacuum. Allow the lysate to filter completely through the filter plate and into the deep well collection plate. Depending on the cell culture density this may take a few minutes. Once the lysate is completely filtered, turn off and release vacuum.
16. Discard filter plate and prepare the manifold by removing the manifold collar. Remove the deep well collection plate containing the filtered lysate from the manifold base and set aside. Place the waste tray in the manifold base. Place the manifold collar over the waste tray so it fits securely on top of the manifold base. Place the binding plate on top of the manifold collar.
17. Add 600 µL of Column Preparation Solution and apply vacuum until all the solution passes through the binding plate. Turn off vacuum.
18. Transfer the filtered lysate from the deep well collection plate to the binding plate.

19. Apply vacuum so that the solution slowly passes through the binding plate. Turn off vacuum.
20. Add 600 µL of Wash Solution 1 and apply vacuum until all the solution passes through the binding plate. Turn off vacuum.
21. Add 600 µL of Wash Solution 2 and apply vacuum until all the solution passes through the binding plate. Turn off vacuum.
22. Remove the binding plate from the vacuum manifold and thoroughly blot the plate on an absorbent pad, making sure all residual Wash Solution 2 is removed.
23. Place the binding plate back on the vacuum manifold, apply maximum vacuum and dry the binding plate for a minimum of 10 minutes.
24. Once the plate is dry, place the binding plate on top of the elution plate.
25. Add 100 µL of Elution Solution directly onto the membrane for each well of the binding plate and centrifuge the plate at 1800 x g for 5 minutes.

shRNAmir inserts preparation

26. shRNAmir inserts were prepared using two approaches:

    Direct approach:

    a. Pool the purified plasmid DNA from the original GIPZ clones.
    b. Cleave the pool DNA with XhoI and EcoRI to release all 96 inserts simultaneously.
    c. Purify the inserts after restriction enzyme digestion using QIAquick gel extraction kit.

    PCR approach:

    a. Amplify each individual shRNAmir insert in 20 µL reactions containing 5 ng of plasmid DNA using the following recipe.

5 x Phusion buffer	4 µL
10 mM dNTPs	0.4 µL
10 □M A XhoI-5'mir30	1 µL
10 □M EcoRI-3'mir30	1 µL
DNA	5 ng
Phusion Hot Start II High-Fidelity DNA polymerase	0.2 µL
H_2O	Up to 13.4 µL
Total	20 µL

    b. Primer sequences with the restriction enzymes sequences underlined:
       XhoI-5'mir30:
       CAGAAGGCTCGAGAAGGTATATTGCTGTTGACAGTGAGCG
       EcoRI-3'mir30:
       CTAAAGTAGCCCCTTGAATTCCGAGGCAGTAGGCA
    c. Amplification protocol.
       a. 94°C – 5 min
       b. 94°C – 30 sec
       c. 54°C – 30 sec
       d. 75°C – 30 sec (b->d 16 cycles)
       e. 75°C – 2 min
    d. Confirm the amplification of each shRNAmir with agarose gel electrophoresis.
    e. Pool the PCR products and digest with DpnI to destroy remaining template plasmid DNA for 1 hour at 37°C.
    f. Digest the pooled PCR products with EcoRI and XhoI to generate sticky ends.
    g. Purify the inserts after restriction enzyme digestion using QIAquick gel extraction kit.

Ligation and transformation

27. Ligate inserts prepared with each method separately into 100 ng of pLMPd-Amt in a 2.5:1 molar ratio.
28. Transform 10% of each ligation into 50 µL of XL10-Gold chemically competent cells.
29. Pick 384 colonies from each transformation.
30. Culture in 96-well plates containing 200 µL of LB medium with 8% of glycerol and 50 µg/mL Ampcillin for 24 hour at 37°C without shaking.
31. Mix the culture and freeze in -80°C.

DNA sequencing and analysis

32. Send the frozen culture to Beckman Genomics for DNA sequencing.
33. Align all reads to a reference file from the 96 shRNA sequences of the original GIPZ library using the BLAST tool.

Supplemental References

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